

From Memory to Belief: A Survey of State Maintenance and Belief Revision in LLM Decision Agents

Jimin Huang^{1,2} Yuyan Wang¹ Xueqing Peng² Sophia Ananiadou¹ Jun'ichi Tsujii³

¹The University of Manchester ²The Fin AI ³AIST, Japan

jimin.huang@postgrad.manchester.ac.uk

Abstract

Language-model agents now plan, call tools, and act over many steps, and the failures that dominate reports from such settings are failures of state rather than of reasoning: the agent loses track of what it has learned, keeps acting on evidence that has since been contradicted, or revises what it believes without changing what it does. Memory mechanisms, uncertainty estimation, and post-training with denser rewards each address one part of this problem, but the object that connects them, a persistent, uncertainty-aware belief about a partially observed world, has no survey of its own. This survey takes the belief state as its organising object and traces its development from 2020 to 2026, from context and memory stores through external Bayesian filters and model-written beliefs to learned world models exposed to the policy. We ask one question of every method, *which object receives credit*, the output, the trajectory, the interaction, or the state, and use it to organise three axes: how the belief is represented and whether it commits to testable predictions, how it is revised and where revision fails, and how learning signals reach it. We then align the fragmented belief-level evaluation literature by construct, catalogue 33 benchmarks, and identify the gap the alignment exposes: no published method yet obtains a signed learning signal for a written belief from the agent's own predictions.

1 Introduction

Language-model agents now plan, call tools, retrieve evidence, and act over horizons of hours or months, and the failures reported from these settings are failures of state rather than of reasoning. A model that solves a problem in one pass loses much of its accuracy when the same information arrives over several turns (Laban et al., 2025); an agent that plans well falls to near-random success once the environment moves while it deliberates (Gonzalez-Pumariega et al., 2025); an agent

Survey (venue)	Organising axis	B	R	S	E
MEMORY					
Zhang et al. (2024b) (arXiv)	design, evaluation, use	○	●	○	○
Hu et al. (2025b) (arXiv)	factual / experiential / working	○	●	○	●
Luo et al. (2026a) (ACL F)	storage → reflection → experience	○	●	○	○
REWARDS AND CREDIT					
Wu (2025) (EMNLP F)	reward model × stage	○	○	○	○
Zheng et al. (2025) (ACL)	data → PRM → use	○	○	○	○
Zhang et al. (2025) (TMLR)	capability × application	●	○	○	○
Zhang (2026) (arXiv)	granularity × method	○	○	●	○
UNCERTAINTY					
Xia et al. (2025) (ACL F)	four estimation avenues	○	○	○	●
Lin et al. (2026b) (arXiv)	what / how / where	●	○	○	●
EVALUATION, HORIZON, DOMAIN					
Yehudai et al. (2026) (ACL F)	capability × domain	○	○	○	○
Chen et al. (2026a) (arXiv)	lifecycle × horizon carrier	●	●	○	○
Xu et al. (2026c) (arXiv)	forecasting architecture	●	○	○	●
Dong et al. (2025b) (EMNLP F)	financial function	○	○	○	○
This survey	credited object	●	●	●	●

Table 1: Neighbouring surveys and the belief state. **B**: belief as an object of its own; **R**: revision and its failure modes; **S**: learning signals that reach the belief; **E**: belief-level evaluation. ● yes, ○ partly, ○ no.

given only observations wins a fraction of the episodes won by one that reads the simulator's latent state (Alaswad et al., 2026); and in markets, where evidence arrives late and partially, stronger backbones do not reliably produce stronger trading agents (Xie et al., 2024; Li et al., 2024). What these failures share is that the task required the agent to hold something between decisions, a belief about what is true, how certain it is, and what would change it, and the agent never formed it, never tested it, or revised it without acting on the revision.

The field has responded along three lines that rarely cite one another. Memory research moved the state from the reasoning trace (Wei et al., 2022b; Yao et al., 2023b) into external stores that persist, reflect, and abstract experience (Park et al., 2023; Shinn et al., 2023; Luo et al., 2026a). Uncertainty research taught models to state confidence (Lin et al., 2022; Kadavath et al., 2022) and extended it from single answers to agent trajectories (Lin et al., 2026b). Post-training moved

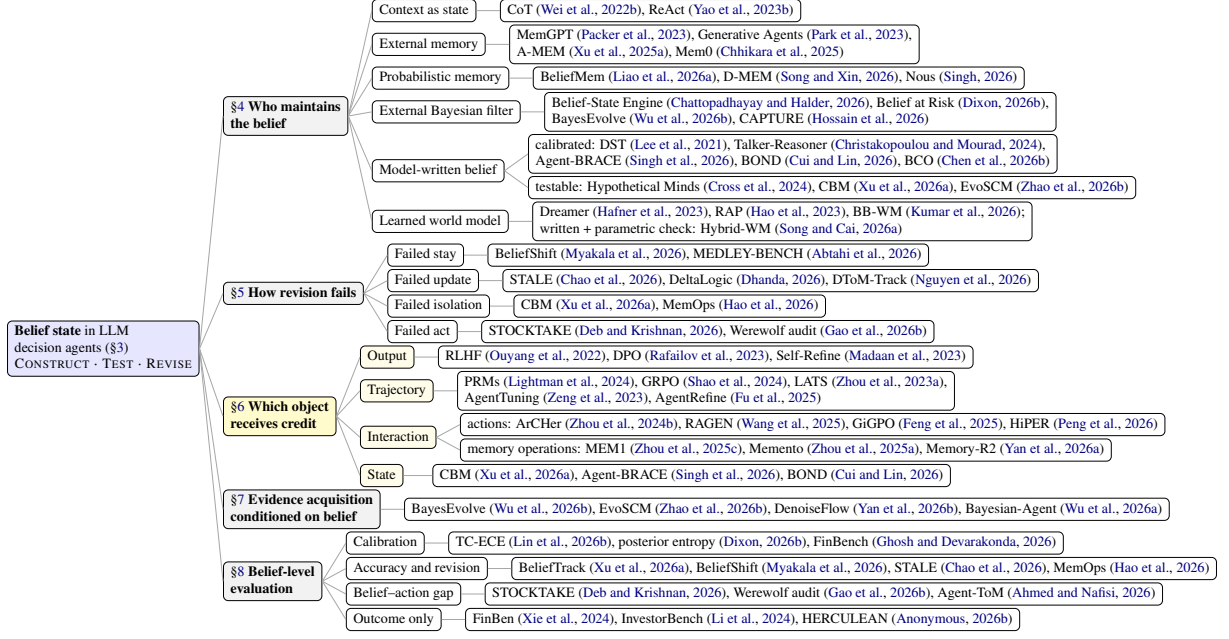


Figure 1: Taxonomy of the survey with representative systems by year. The credited object (yellow) is the axis every method is placed on; Table 12 in the appendix places every screened system.

from rewarding the finished response (Ouyang et al., 2022) to rewarding reasoning steps (Lightman et al., 2024; Zheng et al., 2025) and then actions over several turns of interaction (Zhou et al., 2024b; Zhang et al., 2025; Zhang, 2026). Each line supervises a different thing, what is stored, how confident the answer is, which action earns reward, and none of them supervises the belief itself.

Explicit beliefs are older than the recent attention suggests: dialogue state tracking wrote one in 2021 (Lee et al., 2021), agents wrote beliefs about tasks, opponents, and markets in 2024 (Kim et al., 2024; Cross et al., 2024; Yu et al., 2024), and two 2025 systems learned to overwrite a written state by reinforcement (Zhou et al., 2025c; Yu et al., 2025). Until recently the belief was credited, if at all, by the task outcome. In 2026 alone at least twelve systems gave it a learning signal or oracle of its own,¹ and benchmarks began to score the belief rather than the outcome it led to (Xu et al., 2026a; Deb and Krishnan, 2026; Chao et al., 2026), with disjoint formalisms and metrics and little mutual citation (timeline in Figure 7, appendix).

No existing survey takes this object as its own: memory surveys organise by what is stored, re-

ward and credit surveys stop at the action, uncertainty surveys score an output rather than a state, and evaluation surveys report no belief-level metric (Section 2, Table 1). This survey asks one question of every method: *which object receives credit*. Figure 1 gives the resulting taxonomy, and Figure 2 places the four answers on the observation–belief–action loop: the finished output, the sampled trajectory, the interaction (actions, tool calls, memory operations), or the belief written before the action and tested by evidence returned after it. It cuts across the neighbouring taxonomies rather than replacing them. We cover language-model agents acting over several steps under partial observability, from 2020 to September 2026, collected by a scripted protocol (Appendix A). Section 3 grounds the belief state in POMDP belief and belief-revision theory; Sections 4–8 organise the literature by representation, revision, credited object, evidence acquisition, and evaluation; Section 9 compares the four levels, and Section 10 states the open problems, chief among them that no published method obtains a signed signal for a written belief from the agent’s own predictions. Finance is the running example because delayed, partial evidence is its normal condition; the taxonomy is stated without reference to any domain.

¹Belief-State Engine, Belief at Risk and its validation framework, BayesEvolve, CAPTURE, Agent-BRACE, the Bayesian linguistic forecaster, BOND, CBM, BCO, EvoSCM, and the belief-based world model; Section 4.

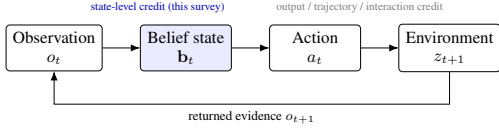


Figure 2: The observation-belief-action loop. Existing post-training credits the action, the trajectory that produced it, or the response text. State-level credit reaches the belief written before the action, using evidence returned after it.

2 Related Work

Table 1 places this survey among thirteen neighbouring ones, none of which takes the belief state as its object. Memory surveys (Zhang et al., 2024b; Hu et al., 2025b; Luo et al., 2026a) organise by what is stored and never ask whether a stored item is believed or what evidence would retire it. Reward and credit surveys (Wu, 2025; Zheng et al., 2025; Zhang et al., 2025; Zhang, 2026) follow the signal down to tokens, steps, and memory operations and stop at the action. Uncertainty surveys (Xia et al., 2025; Lin et al., 2026b) attach a score to an output rather than to a state carried forward. Evaluation, long-horizon, forecasting, and finance surveys (Yehudai et al., 2026; Chen et al., 2026a; Xu et al., 2026c; Dong et al., 2025b) treat belief as one component among many and report no metric that scores it apart from the outcome.

3 Problem Formulation

Partially observed decision-making defines the belief as the distribution over hidden variables that is sufficient for acting: with hidden state z_t , observation o_t , and history H_t , the belief is $\mathbf{b}_t = p(z_t | H_t)$, maintained by a Bayes filter (Åström, 1965; Kaelbling et al., 1998; Thrun et al., 2005). Agentic reinforcement learning casts the language-model agent in the same terms, a POMDP whose observable history is the context, with any explicit state kept in a working memory the model itself writes (Zhang et al., 2025; Sumers et al., 2023). In a ReAct-style agent each turn emits a reasoning segment and then an action (Yao et al., 2023b); the reasoning segment is where the state lives, and because the context only grows, the state is carried forward by being rewritten or re-read rather than by an estimator. We write $\mathbf{b}_t$ for whatever explicit state the span carries before the action a_t , and note that a later observation o_{t+1} can evaluate two different things: the action, and the belief that made

Who maintains the belief	Written	Uncert.	Testable	Auditable	Example
Context as state	●	○	○	○	ReAct
External memory	○	○	○	●	MemGPT
Probabilistic memory	○	●	○	○	Nous
External Bayesian filter	○	●	●	●	Belief-State Engine
Written, calibrated	●	●	○	○	Agent-BRACE, BOND
Written, testable	●	○	●	○	CBM, EvoSCM
Learned world model	○	○	●	○	Dreamer, BB-WM
Written + parametric check	●	●	●	○	Hybrid-WM

Table 2: Representation classes and the properties they trade off. *Written*: the model writes and reads the belief; *Uncert.*: it carries uncertainty; *Testable*: it commits to a falsifiable prediction; *Auditable*: it can be validated against a closed-form oracle. ● yes, ● partly, ○ no.

the action plausible.

Two older literatures say what such a belief must do. Bayesian filtering requires it to predict and to update on the mismatch (Thrun et al., 2005; Friston, 2010); belief-revision theory requires that a rational agent can expand its beliefs with a new claim, revise them when a claim is contradicted, and contract them without discarding the rest (Alchourrón et al., 1985). Read together, they give the three operations by which we place every system: CONSTRUCT, writing a claim and binding observations to it; TEST, committing to a prediction that a returned observation can falsify; and REVISE, changing the active belief without erasing what was believed. A *memory* stores traces and is judged by recall (Luo et al., 2026a); a *world model* predicts the next observation and is judged by prediction error (Li et al., 2026c); a *belief* interprets traces as claims about what is true now, with uncertainty, and is judged by calibration and by how fast it is revised when wrong. A reasonable action under a stale belief, the failure that motivates this survey, is invisible to both neighbours.

4 Belief Representation: Who Maintains the State

Where does the belief live, who writes it, and can it be wrong in a way that later evidence can show? We order representations by *who maintains the belief*, from no one to an explicit estimator, and by whether the belief is *testable*, that is, commits to a prediction that a returned observation can falsify. Table 2 summarises the classes; Figure 6 and Table 12 in the appendix place every system on both axes.

4.1 Context and Memory

The most common design has no representation at all: the reasoning trace is the state (Wei et al., 2022b; Yao et al., 2023b), the belief implicit and its uncertainty unstated. In a text game, agents that read the simulator’s latent state win about 79% of episodes against 11% for agents that read only observations (Alaswad et al., 2026), and on STALE the best model resolves old-versus-new conflicts in 55% of cases (Chao et al., 2026), and programmatic state abstraction alone helps substantially (Bogdanov et al., 2026; He et al., 2026a).

Memory-augmented agents moved the state into a store retrieved into context (Packer et al., 2023; Park et al., 2023; Zhong et al., 2024; Xu et al., 2025a), the evolution that Luo et al. (2026a) survey. Two limits matter for belief: the stored unit is a trace rather than a claim with a probability, and an update is an operation on the store rather than on a hypothesis, so answer accuracy hides failures in the update and forget operations themselves (Hao et al., 2026). Structured stores narrow the first limit with schemas, graphs, or bitemporal transitions (Xu and Fekri, 2026; Luo et al., 2026b; Liu et al., 2026b; Xu, 2026; Ning et al., 2026); probabilistic stores narrow the second by attaching a number to each claim (Liao et al., 2026a; Song and Xin, 2026; Yang et al., 2026b). Even closed-form Bayesian updates per stored claim barely beat last-write-wins on LoCoMo (Singh, 2026; Srivastava, 2026).

4.2 External Bayesian Filters

A different line takes the belief out of the model. The model maps evidence to a likelihood over a fixed set of latent regimes, hypotheses, or user states, and an explicit filter maintains the posterior, whose entropy and drift can then be reported as model risk (Chattopadhyay and Halder, 2026; Dixon, 2026b,c,a; Wu et al., 2026b; Hossain et al., 2026; Soru, 2026). These are the only beliefs in the survey validated against a closed-form oracle, and they are auditable by construction. The price is a hypothesis space fixed in advance: the filter cannot introduce a factor it was not given.

4.3 Model-Written Beliefs

The third line keeps the belief in context but gives it structure the model writes and reads. Dialogue state tracking wrote a slot-value belief in 2021 (Lee et al., 2021), verbalised confidence attached

a probability to an answer (Lin et al., 2022; Kada-vath et al., 2022), 2024 agents wrote beliefs about tasks, opponents, and markets (Kim et al., 2024; Christakopoulou and Mourad, 2024; Cross et al., 2024; Yu et al., 2024), and in 2025 MEM1 and MemAgent learned to overwrite a fixed written state each turn (Zhou et al., 2025c; Yu et al., 2025). The newest systems add uncertainty and verification to the written object, from certainty tags and Bayesian-teacher posteriors to symbolic verification and populations of causal models (Singh et al., 2026; Murphy, 2026; Cui and Lin, 2026; Xu et al., 2026a; Chen et al., 2026b; Seo et al., 2026; Giovannelli and Kent, 2026; Zhao et al., 2026b).

Testability separates them, and we define it operationally: a written belief is testable if it contains a statement about a future observation with a horizon, so that the returned observation yields a signed error for that statement. By this definition most written beliefs are calibrated but not testable: they state confidence or revise the belief but never name the observation that would falsify it. Hypothetical Minds, CBM, and EvoSCM do, and Section 6 turns on this distinction.

4.4 Learned World Models

The fourth line learns the dynamics and exposes the resulting belief to the policy: recurrent latent states (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2023), the language model as its own world model for tree search (Hao et al., 2023; Zhao et al., 2023b; Zhou et al., 2023a), LLM-proposed POMDPs refined to a belief-based likelihood (Six et al., 2026), and a world-model belief the policy can query (Kumar et al., 2026; Li et al., 2026c; Bagaria et al., 2026). World-model beliefs are testable by construction but, unless written back, cannot be reasoned about in language. Two 2026 systems close that gap by using the learned model as a check on a written belief rather than a replacement: a parametric consistency gate that cuts the hallucinated-state rate from 0.176 to 0.035 (Song and Cai, 2026a), and semantic rules distilled from prediction–observation mismatches (Zhang et al., 2026c).

Insight 1. Structure is not testability. Memory design has made the belief steadily more structured without making it falsifiable, and the representations that can be falsified, filters and world models, cannot be reasoned about in language. The belief the field needs is written and testable at once; no peer-reviewed system has one yet.

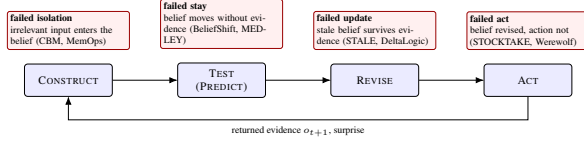


Figure 3: Where the four failure modes of Section 5 occur on the construct–test–revise–act cycle. Only a testable belief lets returned evidence be routed to the operation that failed rather than to the action.

Failure mode	Definition	Measured by	Headline result
Failed stay	belief moves without evidence	BeliefShift, MEDLEY-BENCH	consensus sensitivity from near 0 to large across 35 models
Failed update	belief survives contradicting evidence	DeltaLogic, DToM-Track, STALE	66.7% on a conclusion, 46.7% on whether an edit changes it
Failed isolation	irrelevant input enters the belief	BeliefTrack, MemOps	belief-state reward cuts failures by 71%
Failed act	belief revised, action unchanged	STOCKTAKE, Werewolf audit	34–43% of correctly diagnosed weeks end in the wrong action

Table 3: Four failure modes of belief revision, the benchmarks that isolate each, and their headline results (Myakala et al., 2026; Abtahi et al., 2026; Dhanda, 2026; Nguyen et al., 2026; Chao et al., 2026; Xu et al., 2026a; Hao et al., 2026; Deb and Krishnan, 2026; Gao et al., 2026b).

5 Belief Revision and Its Failure Modes

Bayesian filtering and active inference agree on how a belief should change: it predicts, evidence arrives, and the mismatch drives the update (Friston, 2010; Friston et al., 2017); factor graphs then assign the error to the factors that produced the prediction (Kschischang et al., 2001; Loeliger, 2004). The language-model literature imported the vocabulary but rarely the routing: feedback is appended to the context or folded into self-critique loops that revise an output, a plan, or a trajectory (Huang et al., 2022b; Madaan et al., 2023; Chen et al., 2023b; Kim et al., 2023; Wang et al., 2023b).

5.1 Four Failure Modes

Contextual Belief Management names three ways a belief can fail on new input, a *failed stay* in which the belief moves without evidence, a *failed update* in which it survives evidence against it, and a *failed isolation* in which irrelevant input contaminates it (Xu et al., 2026a). Three independent studies document a fourth, the *failed act*, in which the belief is revised but the action is not, so that a belief-level oracle finds the diagnosis right while

the action stays wrong (Deb and Krishnan, 2026; Gao et al., 2026b). Table 3 lists the benchmark that isolates each mode; their placement on the construct–test–revise–act cycle (Figure 3) is ours, and each names the operation whose signal should receive the error.

5.2 Mechanisms of Revision

Failed update is now a benchmark topic in its own right (Chao et al., 2026; Sun and He, 2026; Nakayashiki, 2026; Yang et al., 2026a); in finance it is a stale factor, visible as drift in the agent’s planning embeddings before the trading loss appears (Xue, 2026). The mechanisms that respond fall into three families by what triggers them. The first is triggered by *contradiction* among stored claims and resolves it with an operator algebra, first-class update and forget operations, or super-session contracts that keep the superseded claim visible (Wang, 2026; Hao et al., 2026; Xu, 2026). The second is triggered *before commitment*, by self-audit, a parametric veto, semantic entropy, or resampling (Yuan et al., 2026; Song and Cai, 2026a; Yan et al., 2026b; Wang et al., 2026b); these see no future observation and so cannot catch a belief that is consistent and wrong. The third is triggered by *surprise*: reward-prediction error routes writes to a fast or a slow path (Song and Xin, 2026), and prediction–observation mismatches become semantic rules (Zhang et al., 2026c).

Routing the error to the claim that produced the prediction, the factor-graph step with which this section opened, is what none of the surveyed systems implements in the model’s own context.

Insight 2. Revision is triggered, not routed. Contradiction, self-audit, and surprise decide *when* a belief changes; none decides *which claim* the error belongs to. Until the error is routed to the claim that made the prediction, agents will keep repairing memories while beliefs stay wrong, and failed act will remain the unmeasured mode.

6 Learning Signals by Credited Object

Existing surveys organise post-training by where the reward comes from, where the trajectories come from, and how dense the signal is (Wu, 2025; Zhang, 2026; Zheng et al., 2025). We hold those fixed and vary one thing, the object that receives the gradient (Figure 4, Table 4). The criterion is operational: a method is at the *state level* if the quantity that receives a gradient is a claim about the environment and its sign depends on an observation returned after the action. A method that rewards a memory write regardless of whether the

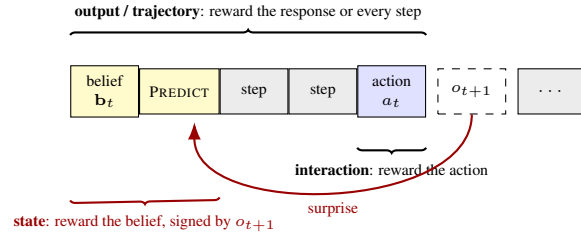


Figure 4: The credited object on one span of a trajectory. Output- and trajectory-level credit reaches text generated from the context; interaction-level credit reaches the action; state-level credit reaches the written belief, and its sign comes from the observation returned after the action.

Level	Credited object	Signed by o_{t+1} ?	Representative
Output	finished response	no	RLHF, DPO
Trajectory	sampled steps	no	PRMs, GRPO, LATS
Interaction	actions, tool calls	no (task reward)	ArCher, RAGEN, HiPER
Interaction	memory operations	no (operation reward)	MEM1, Memento, Memory-R2
State	written belief	closed-world oracle	CBM
State	claim + certainty	calibration target	Agent-BRACE
State	posterior tag	Bayesian teacher	BOND
State	written belief	<i>own prediction</i>	(none published)

Table 4: Learning signals by credited object and where the sign of the signal comes from (Section 6); the full version with feedback source, granularity, and evidence level is Table 10 in the appendix.

written claim was true is at the interaction level, however dense its credit.

6.1 Output and Trajectory

Supervised fine-tuning and preference optimisation credit the completed response (Ouyang et al., 2022; Wei et al., 2022a; Rafailov et al., 2023), and self-consistency and self-improvement select the responses to train on (Wang et al., 2022; Huang et al., 2022a). Trajectory-level credit began by imitating and searching over agent trajectories (Chen et al., 2023a; Zeng et al., 2023; Xi et al., 2024; Zhou et al., 2023a; Putta et al., 2024; Fu et al., 2025), then credited sampled steps directly through PPO, GRPO, and process reward models (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024; Zheng et al., 2025; Zhang et al., 2026e; Tan et al., 2026).

6.2 Interaction

Agentic reinforcement learning credits actions over several turns (Zhou et al., 2024b; Qi et al., 2024; Wang et al., 2025; Chen et al., 2025a; Zhou et al., 2025b; Feng et al., 2025; Dong et al., 2025a), and the open question is how to make that credit denser without making it wrong, by hierarchical advantages, counterfactual estimators over

Acquisition conditioned on	conditioned on	Systems	Belief object
fixed heuristic		agentic RAG (Gao et al., 2024)	none
task-level POMDP		CGDP (Kausik et al., 2026), MedExAgent (Gao et al., 2026a)	searcher, patient state
uncertainty of an explicit belief		BayesEvolve (Wu et al., 2026b), EvoSCM (Zhao et al., 2026b), DenoiseFlow (Yan et al., 2026b)	hypothesis posterior, causal population, step entropy
belief over own resources		Bayesian-Agent (Wu et al., 2026a), REDEREF (Hosseini et al., 2026), CMI (Srivastava, 2026)	skill, delegate, memory reliability
information gain as reward		PiCA (Liu et al., 2026a), TIPS (Xie et al., 2026)	implicit
written testable belief		<i>none found</i>	—

Table 5: Evidence-acquisition decisions by what they are conditioned on.

shared states, potential-based shaping, or verifier-structured credit (Peng et al., 2026; Wang et al., 2026a; Diao et al., 2026; Wang et al., 2026f; Ren et al., 2026; Liao et al., 2026b; He et al., 2026b; Liu et al., 2026a; Xie et al., 2026; Zheng et al., 2026; Zhang et al., 2026b; Li et al., 2026a; Wang et al., 2026g; Guo et al., 2026; Qi et al., 2026; Zhao et al., 2026a; Lin et al., 2026a); denser is not free (Modecrua et al., 2026; Pu et al., 2026). The nearest this level comes to the state is a group that credits the write to a memory, from a rewritten case bank to an RL-trained overwrite of a fixed written state (Zhou et al., 2025a,c; Yu et al., 2025; Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026; Wang and Jiang, 2026; Hu et al., 2026). By the criterion above these remain interaction-level: they credit the *operation*, so a well-executed update to a wrong claim is still rewarded.

6.3 State

Three published methods meet the criterion. CBM trains with belief-state rewards from symbolic verification and cuts belief-management failures by about 71% (Xu et al., 2026a). Agent-BRACE trains belief model and policy jointly with verbalised certainty as the belief’s output (Singh et al., 2026). BOND distils a Bayesian teacher’s posterior into an 8B student whose Brier score beats a 70B prompted baseline (Cui and Lin, 2026).

Insight 3. Credit has been getting denser, not deeper. Agentic reinforcement learning has refined which action to prefer; no method changed what the action is conditioned on. The two axes are orthogonal, and the first system to combine them will need a sign for its belief that the environment, not an oracle, supplies.

	calibration	accuracy	revision	belief-action	outcome
UQ taxonomy (Lin et al., 2026b) A	•	◦	◦	◦	◦
Agent-BRACE (Singh et al., 2026) A	•	◦	◦	◦	◦
Belief-State Engine (Chattopadhyay and Halder, 2026) A	•	◦	◦	•	◦
Belief at Risk (Dixon, 2026b) A	•	◦	◦	◦	◦
FinBench (Ghosh and Devarakonda, 2026) A	•	◦	◦	◦	•
BeliefTrack (Xu et al., 2026a) A	◦	•	•	◦	◦
DeltaLogic (Dhanda, 2026) A	◦	•	•	◦	◦
DToM-Track (Nguyen et al., 2026) A	◦	•	•	◦	◦
BeliefShift (Myakala et al., 2026) A	◦	◦	•	◦	◦
STALE (Chao et al., 2026) A	◦	◦	•	•	◦
MemOps (Hao et al., 2026) A	◦	◦	•	◦	◦
STOCKTAKE (Deb and Krishnan, 2026) A	◦	•	◦	•	•
Werewolf audit (Gao et al., 2026b) A	◦	◦	◦	•	•
HERCULEAN (Anonymous, 2026b) A	◦	◦	◦	◦	•
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024) P	◦	◦	◦	◦	•

Figure 5: Which belief-level construct each evaluation paper reports. Filled dots mark a reported metric. P = peer-reviewed venue, A = arXiv preprint at the time of writing. No two papers report the same metric for the same construct, which is the fragmentation this section addresses.

7 Belief-Guided Evidence Acquisition

A testable belief tells the agent what evidence would change it, which turns retrieval, tool use, and routing from fixed pipeline stages into decisions (Table 5). Agentic search (Kausik et al., 2026) and clinical diagnosis (Gao et al., 2026a) have been framed as POMDPs whose actions include what to observe next, and two finance simulators show why it matters: failing agents spend 62% of their observation budget re-reading internal state, and failures trace to incomplete acquisition rather than to reasoning (Han et al., 2026; Zhang et al., 2026a).

The methods that condition acquisition on a belief differ in what the belief is over (Table 5). Agentic retrieval first let the model decide when to retrieve (Gao et al., 2024); recent systems condition that decision on measured uncertainty about the task (Wu et al., 2026b; Zhao et al., 2026b; Xu and Fekri, 2026; Yan et al., 2026b), make information gain the training signal itself (Liu et al., 2026a; Xie et al., 2026), or keep a belief over the agent’s own skills, delegates, or memories (Wu et al., 2026a; Hosseini et al., 2026; Srivastava, 2026). Financial evidence adds a constraint, since much of it cannot leave its silo; knowledge harnesses and federated adaptation are the compliance-aware forms of integration (Borjigin et al., 2026; Anonymous, 2026a).

Insight 4. Acquisition is a belief problem. Agents lose by re-reading what they know rather than seeking what would change it. A testable belief names that evidence, so evidence acquisition will move from a fixed retrieval stage to a consequence of the belief.

8 Evaluating Belief Apart from Outcome

A realised trade or a correct answer can follow from a correct belief, from a wrong belief that did not matter, or from a belief revised too late, and outcome metrics cannot separate the three; this is why stronger backbones do not yield stronger trading agents and the benchmarks cannot say why (Xie et al., 2024; Li et al., 2024). Earlier agent benchmarks score task success or recall (Shridhar et al., 2020; Liu et al., 2023; Maharana et al., 2024; Yao et al., 2024); recent work responded with belief-level metrics, each paper defining its own. Figure 5 aligns them by construct, and Table 11 in the appendix catalogues 33 benchmarks by whether the evaluator can observe the latent state.

8.1 Constructs

Calibration over a trajectory has the most statistics and the least comparability: checkpoint calibration error, episode-level calibration, posterior entropy, and proper scoring rules on time-gated forecasts express one construct in four ways, none computed on another paper’s system (Lin et al., 2026b; Singh et al., 2026; Chattopadhyay and Halder, 2026; Dixon, 2026b; Ghosh and Devarakonda, 2026). *Belief accuracy and revision* are measured against a known answer in a closed world, after a minimal premise edit, and across recall, inference, and change detection (Xu et al., 2026a; Dhanda, 2026; Nguyen et al., 2026), and against human annotation (Myakala et al., 2026; Chao et al., 2026; Hao et al., 2026; Yang et al., 2026a) and, for beliefs about other agents, by theory-of-mind benchmarks (Bawatneh et al., 2026; Wang et al., 2026c). The *belief-action gap* is the youngest construct: STOCKTAKE scores the agent against an exact Bayes filter on the same observations (Deb and Krishnan, 2026), the Werewolf audit reports action-belief consistency (Gao et al., 2026b), and Agent-ToM monitors an agent’s beliefs from its trajectory (Ahmed and Nafisi, 2026).

8.2 Where the Latent State Comes From

Every construct beyond calibration needs a latent state to compare against. Controlled environments provide it (Deb and Krishnan, 2026; Xu et al., 2026a; Alaswad et al., 2026); long-horizon simulators know it but report belief only through proxies (He et al., 2026a; Zhang et al., 2026a; Han et al.,

Level	Sign of the signal from	Env.	No oracle	Belief	Act gap	Maturity
Output	humans, verifiers	○	●	○	○	●
Trajectory	verifier, PRM	●	●	○	○	●
Interaction (actions)	task reward	●	●	○	○	●
Interaction (memory ops)	task reward, teacher	●	●	○	○	○
State (oracle or teacher)	closed world, teacher, target	○	○	●	●	○
State (own prediction)	returned observation	●	●	●	●	○

Table 6: The four credit levels compared. *Env.*: the sign of the signal comes from the environment; *No oracle*: no external source of truth is needed; *Belief*: the written belief receives the gradient; *Act gap*: belief error and action error are separated; *Maturity*: ● many systems, ● a few, ○ none published.

2026); historical replay provides neither, so workflow benchmarks can report a belief failure only at the outcome level (Anonymous, 2026b).

Insight 5. Belief cannot be scored by outcome. Five constructs, fifteen metrics, no cross-paper comparison; only the fair oracle scores belief and action on the same episodes. Progress on belief will be measurable only where the latent state is known, so controlled environments, not replay, are where the next results should come from.

9 Discussion

Table 6 compares the four credit levels along the properties the preceding sections exposed; three observations follow.

The sign is the scarce resource. Output and trajectory levels take the sign of their signal from humans or verifiers, the interaction level from the task reward, and the state level, in all three published cases, from an oracle or teacher the environment does not supply. Denser credit multiplies a sign the environment already gives; state credit needs a sign for a claim, and without an oracle that sign can only come from the agent’s own prediction settled against the next observation. That row is empty.

Auditability and expressiveness pull apart. Filters are validated in closed form but cannot introduce a factor they were not given; written beliefs can name a new factor but are rarely testable; world models are testable but mute. The field is converging on hybrids that write the belief and check it parametrically (Song and Cai, 2026a; Zhang et al., 2026c; Kumar et al., 2026).

The belief–action gap is a training problem. Wherever belief and action are scored on the same episodes, prompting and oracles improve the belief while action–belief consistency barely moves (Deb and Krishnan, 2026; Gao et al., 2026b); the one system that moves both trains belief and policy jointly (Singh et al., 2026). A policy conditioned

on a stale belief does not change when the belief is corrected in context, so closing the gap requires the state-level and interaction-level signals of Table 6 in one objective, not a better belief alone.

10 Open Problems

Self-supervised testability. The state-level methods need an external source of truth, and the surprise-triggered mechanisms of Section 5 have the ingredient that would remove that need but attach the error to a memory entry rather than to the written claim. Combining the two, so that the agent’s own matured predictions supply the signed signal for its written belief, is the most direct open problem this survey identifies. Which evidence horizon to commit to (Ghosh and Devarakonda, 2026; Zhao et al., 2026b) and how much superseded belief must stay visible (Singh et al., 2026; Diao et al., 2026) are its open parameters.

Latent beliefs without ground truth. Every construct beyond calibration needs an oracle, a closed world, or annotation; historical data has none. Filter-recovered latents and representation drift are candidate substitutes, unvalidated against each other (Dixon, 2026b; Xue, 2026); a benchmark pairing a controlled environment with a replay of the same instrument would settle whether belief-level gains transfer.

Belief without action, and belief under governance. Revising the belief does not revise a policy conditioned on the old one, and action–belief consistency sits far below belief accuracy wherever both are measured (Gao et al., 2026b; Deb and Krishnan, 2026). Regulated domains add that the belief must be auditable, which today only fixed-space filters provide; LLM-proposed spaces (Six et al., 2026; Soru, 2026; Chen et al., 2026b) are the candidate bridge (Dixon, 2026c).

11 Conclusion

This survey asked one question of the literature on language-model decision agents: which object receives credit. Organising 179 papers by the answer exposes a field that has moved the belief from the trace into stores, filters, and world models, learned to make credit denser at every level below the state, and built fifteen ways to measure belief without a shared one. Three findings carry across the axes. The representations that can be falsified are not the

ones the agent can reason about; the mechanisms that revise beliefs do not route the error to the claim that produced it; and the three methods that credit the belief at all borrow the sign of their signal from an oracle the environment does not supply. The gap they leave is precise, and we argue it is the next step: an agent whose own matured predictions supply the signed signal for its written belief, evaluated where the latent state is known. The taxonomy, screening record, and benchmark catalogue are released to make that step measurable.²

Limitations

This survey covers work indexed by OpenAlex and arXiv from January 2020 to 15 September 2026 and retrieved by the keyword protocol in Appendix A; papers that describe belief maintenance without using any of our search terms may be missed. Our taxonomy assigns each method to one representation class and one credit level, although several systems combine classes. The survey reports numbers as stated by the cited papers and does not re-run any method; headline figures from preprints may change in later versions. Finance is used as the running example because delayed and partially observed evidence is the norm there; the taxonomy itself is domain-agnostic, but we have not validated it against embodied or dialogue agents in the same depth.

References

- Farhad Abtahi, Abdolamir Karbalaie, Eduardo Illueca-Fernandez, and Fernando Seoane. 2026. [Medley-bench: Benchmarking behavioural metacognition and belief revision under social pressure in large language models](#). *Preprint*, arXiv:2604.16009. ArXiv:2604.16009.
- Nesreen K. Ahmed and Nima Nafisi. 2026. [Agent-tom: Learning to monitor autonomous llm agents via theory-of-mind reasoning](#). *Preprint*, arXiv:2605.24216. ArXiv:2605.24216.
- Feisal Alaswad, Batoul Aljaddouh, Maher Alrahhah, Poovammal E, and Talal Bonny. 2026. [Why we need world models for agi: Where llms fail and how world models may outperform](#). *Preprint*, arXiv:2605.23972. ArXiv:2605.23972.
- Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. 1985. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530.
- Amez Amanj Ali and Kuo-Kun Tseng. 2026. [Reward-driven llm agent workflows: Synthesizing pomdp routing and self-correction for autonomous decision-making](#). *Preprint*, arXiv:2607.17038. ArXiv:2607.17038.
- Anonymous. 2026a. [Concordia: Scorer-guided federated LLM adaptation under compliance constraints](#). *Preprint*, arXiv:2605.09855.
- Anonymous. 2026b. [HERCULEAN: An agentic benchmark for financial intelligence](#). *Preprint*, arXiv:2605.14355.
- Karl Johan Åström. 1965. Optimal control of Markov processes with incomplete state information. *Journal of Mathematical Analysis and Applications*, 10(1):174–205.
- Vaidehi Bagaria, Bijo Sebastian, and Nirav Kumar Patel. 2026. [Recursive belief vision language action models](#). *Preprint*, arXiv:2602.20659. ArXiv:2602.20659.
- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. [tau2-bench: Evaluating conversational agents in a dual-control environment](#). *Preprint*, arXiv:2506.07982. ArXiv:2506.07982.
- Adam Bawatneh, Sagar Sapkota, Amrit Singh Bedi, Santu Karmaker, and Mubarak Shah. 2026. [Omnitom: Benchmarking theory of mind in llms via explicit belief modeling](#). *Preprint*, arXiv:2605.26322. ArXiv:2605.26322.
- Igor Bogdanov, Chung-Horng Lung, Thomas Kunz, Jie Gao, Adrian Taylor, and Marzia Zaman. 2026. [Context, reasoning, and hierarchy: A cost-performance study of compound llm agent design in an adversarial pomdp](#). *Preprint*, arXiv:2605.16205. ArXiv:2605.16205.
- Ailiya Borjigin, Igor Stadnyk, Ben Bilski, Maksym Chikita, Dmytro Korylenko, Sofia Pidturkina, and Julia Stadnyk. 2026. [Absorbing complexity: An interaction-native knowledge harness for financial llm agents](#). *Preprint*, arXiv:2606.01886. ArXiv:2606.01886.
- Hanxiang Chao, Yihan Bai, Rui Sheng, Tianle Li, and Yushi Sun. 2026. [Stale: Can llm agents know when their memories are no longer valid?](#) *Preprint*, arXiv:2605.06527. ArXiv:2605.06527.
- Arnab Chattopadhyay and Debdipta Halder. 2026. [Belief-state engine: Augmenting llms for principled planning under partial observability](#). *Preprint*, arXiv:2609.10036. ArXiv:2609.10036.
- Baian Chen, Chang Shu, Ehsan Shareghi, Nigel Collier, Karthik Narasimhan, and Shunyu Yao. 2023a. [Fire-act: Toward language agent fine-tuning](#). *Preprint*, arXiv:2310.05915. ArXiv:2310.05915.

²<https://github.com/jiminHuang/belief-state-survey>

- Kevin Chen, Marco Cusumano-Towner, Brody Huval, Aleksei Petrenko, Jackson Hamburger, and Vladlen Koltun. 2025a. [Reinforcement learning for long-horizon interactive llm agents](#). *Preprint*, arXiv:2502.01600. ArXiv:2502.01600.
- Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. 2021. Decision transformer: Reinforcement learning via sequence modeling. In *NeurIPS 2021*. ArXiv:2106.01345.
- Mingguang Chen, Licheng Wang, and Bo Qu. 2026a. [The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon llm agents](#). *Preprint*, arXiv:2608.06663. ArXiv:2608.06663.
- Mingyang Chen, Linzhuang Sun, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z. Pan, Wen Zhang, Huajun Chen, Fan Yang, Zenan Zhou, and Weipeng Chen. 2025b. [Research: Learning to reason with search for llms via reinforcement learning](#). In *NeurIPS 2025*. ArXiv:2503.19470.
- Xinyun Chen, Maxwell Lin, and Denny Zhou. 2023b. [Teaching large language models to self-debug](#). In *ICLR 2024*. ArXiv:2304.05128.
- Yuhan Chen, Zhihua Tian, Mahavir Dabas, Charith Peris, Rahul Gupta, Ming Jin, Feiyang Kang, Siyuan Zhang, Nan Wang, and Ruoxi Jia. 2026b. [Belief-calibrated optimization: An explicit world model for agentic optimization](#). *Preprint*, arXiv:2609.01861. ArXiv:2609.01861.
- Mingyue Cheng, Shuo Yu, Daoyu Wang, Qingchuan Li, Xiaoyu Tao, Jie Ouyang, Yucong Luo, Yitong Zhou, Qi Liu, and Enhong Chen. 2025. [Agent-rl: A unified and modular framework for agentic reinforcement learning](#). *Preprint*, arXiv:2511.14460. ArXiv:2511.14460.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. [Mem0: Building production-ready ai agents with scalable long-term memory](#). *Preprint*, arXiv:2504.19413. ArXiv:2504.19413.
- Konstantina Christakopoulou and Shibl Mourad. 2024. [Agents thinking fast and slow: A talker-reasoner architecture](#). *Preprint*, arXiv:2410.08328. ArXiv:2410.08328.
- Logan Cross, Violet Xiang, Agam Bhatia, Daniel LK Yamins, and Nick Haber. 2024. [Hypothetical minds: Scaffolding theory of mind for multi-agent tasks with large language models](#). *Preprint*, arXiv:2407.07086. ArXiv:2407.07086.
- Zongqi Cui and Baihan Lin. 2026. [Distilling bayesian belief states into language models for auditable negotiation](#). *Preprint*, arXiv:2605.04507. ArXiv:2605.04507.
- Sagar Deb and Ashwanth Krishnan. 2026. [Stock-take: Measuring the gap between perception and action in llm agents with a fair oracle](#). *Preprint*, arXiv:2607.13618. ArXiv:2607.13618.
- Amit Dhanda. 2026. [Deltalogic: Minimal premise edits reveal belief-revision failures in logical reasoning models](#). *Preprint*, arXiv:2604.02733. ArXiv:2604.02733.
- Juncheng Diao, Zhicong Lu, Peiguang Li, Yongwei Zhou, Changyuan Tian, Qingbin Li, Rongxiang Weng, Jingang Wang, and Xunliang Cai. 2026. [Hipif: Hierarchical planning and information folding for long-horizon llm agent learning](#). *Preprint*, arXiv:2606.10507. ArXiv:2606.10507.
- Matthew Francis Dixon. 2026a. [Adaptive ai delegation under uncertainty: A bayesian governance policy for sequential decision authority](#). *Preprint*, arXiv:2606.29406. ArXiv:2606.29406.
- Matthew Francis Dixon. 2026b. [Belief at risk: Quantifying agentic ai model risk with llm-inferred bayesian state filters](#). *Preprint*, arXiv:2606.15473. ArXiv:2606.15473.
- Matthew Francis Dixon. 2026c. [Model validation of agentic ai systems: A pomdp-based framework for belief-state, forecast, and policy validation](#). *Preprint*, arXiv:2606.17383. ArXiv:2606.17383.
- Guanting Dong, Hangyu Mao, Kai Ma, Licheng Bao, Yifei Chen, Zhongyuan Wang, Zhongxia Chen, Jiazhen Du, Huiyang Wang, Fuzheng Zhang, Guorui Zhou, Yutao Zhu, Ji-Rong Wen, and Zhicheng Dou. 2025a. [Agentic reinforced policy optimization](#). *Preprint*, arXiv:2507.19849. ArXiv:2507.19849.
- Yifei Dong, Fengyi Wu, Kunlin Zhang, Yilong Dai, Sanjian Zhang, Wanghao Ye, Sihan Chen, and Zhi-Qi Cheng. 2025b. Large language model agents in finance: A survey bridging research, practice, and real-world deployment. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.
- Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. 2025. [Group-in-group policy optimization for llm agent training](#). In *NeurIPS 2025*. ArXiv:2505.10978.
- Karl Friston. 2010. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.
- Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. 2017. Active inference: a process theory. *Neural Computation*, 29(1):1–49.
- Dayuan Fu, Keqing He, Yejie Wang, Wentao Hong, Zhuoma Gongque, Weihao Zeng, Wei Wang, Jingang Wang, Xunliang Cai, and Weiran Xu. 2025. [Agentrefine: Enhancing agent generalization through refinement tuning](#). In *ICLR 2025*. ArXiv:2501.01702.

- Yicheng Gao, Xiaolin Zhou, Yahan Li, Yue Zhao, and Ruishan Liu. 2026a. [Medexagent: Training llm agents to ask, examine, and diagnose in noisy clinical environments](#). *Preprint*, arXiv:2605.07058. ArXiv:2605.07058.
- Yuan Gao, Jiangyi Yang, Yao Zhao, and Yichi Zhang. 2026b. [Auditing belief-conditioned llm agents in hidden-information social deduction games](#). *Preprint*, arXiv:2607.10814. ArXiv:2607.10814.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.
- Luay Gharzeddine and Samer Saab. 2026. [Complete cyclic subtask graphs for tool-using llm agents: Flexibility, cost, and bottlenecks in long-horizon workflows](#). *Preprint*, arXiv:2604.22820. ArXiv:2604.22820.
- Rishab Ghosh and Vinay Devarakonda. 2026. [Fin-bench: Time-gated calibration and uncertainty benchmarking for agentic financial forecasting](#). *Preprint*, arXiv:2607.16229. ArXiv:2607.16229.
- Tommaso Giovannelli and Griffin D. Kent. 2026. [Chal: Council of hierarchical agentic language](#). *Preprint*, arXiv:2605.12718. ArXiv:2605.12718.
- Gonzalo Gonzalez-Pumariaga, Leong Su Yean, Neha Sunkara, and Sanjiban Choudhury. 2025. Robotouille: An asynchronous planning benchmark for LLM agents. In *International Conference on Learning Representations*.
- Yucan Guo, Xiaohan Wang, Miao Su, Saiping Guan, Zhongni Hou, Jiajun Chai, Wei Lin, Guojun Yin, Xiaolong Jin, Jiafeng Guo, and Xueqi Cheng. 2026. [Hidifftr: Hierarchical difficulty-aware policy optimization for multi-turn tool-integrated reasoning](#). *Preprint*, arXiv:2608.21863. ArXiv:2608.21863.
- David Ha and Jürgen Schmidhuber. 2018. World models. In *Advances in Neural Information Processing Systems*.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. 2019. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. 2023. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*.
- Yi Han, Yan Wang, Lingfei Qian, Haohang Li, Yupeng Cao, Yueru He, Xueqing Peng, Nanhan Shen, Yitao Xu, Yankai Chen, Dongji Feng, Jimin Huang, Xue Liu, Jian-Yun Nie, and Sophia Ananiadou. 2026. [Can llm agents be cfos? benchmarking long-horizon resource allocation in an uncertain enterprise environment](#). *Preprint*, arXiv:2603.23638. ArXiv:2603.23638.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *EMNLP 2023*. ArXiv:2305.14992.
- Xixuan Hao, Zeyu Zhang, Zehao Lin, Yihang Sun, Ziliang Guo, Xichong Zhang, Yuxuan Liang, Feiyu Xiong, and Zhiyu Li. 2026. [Memops: Benchmarking lifecycle memory operations in long-horizon conversations](#). *Preprint*, arXiv:2607.12893. ArXiv:2607.12893.
- Muyu He, Adit Jain, Anand Kumar, Vincent Tu, Soumyadeep Bakshi, Sachin Patro, and Nazneen Rajani. 2026a. [Yc-bench: Benchmarking ai agents for long-term planning and consistent execution](#). *Preprint*, arXiv:2604.01212. ArXiv:2604.01212.
- Zhida He, Xiaoyu Wen, Han Qi, Ziyuan Zhou, Peng Yu, Xingcheng Xu, Dongrui Liu, Xia Hu, Chaochao Lu, and Qiaosheng Zhang. 2026b. [Not all turns matter: Credit assignment for multi-turn jailbreaking](#). *Preprint*, arXiv:2605.08778. ArXiv:2605.08778.
- S M Asif Hossain, Ruksat Khan Shayoni, and Md Kishor Morol. 2026. [Capture: Disentangling preference drift from memory poisoning in personalized llm agents](#). *Preprint*, arXiv:2609.02265. ArXiv:2609.02265.
- Mohammad Parsa Hosseini, Ankit Shah, Saiyra Qureshi, Alex Huang, Connie Miao, and Wei Wei. 2026. [Training-free agentic ai: Probabilistic control and coordination in multi-agent llm systems](#). *Preprint*, arXiv:2603.13256. ArXiv:2603.13256.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. 2025a. [Evaluating memory in llm agents via incremental multi-turn interactions](#). *Preprint*, arXiv:2507.05257. ArXiv:2507.05257.
- Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, Boyang Liu, Fangyi Zhu, Jiahang Lin, Honglin Guo, Shihan Dou, Zhiheng Xi, et al. 2025b. Memory in the age of AI agents. *arXiv preprint arXiv:2512.13564*.
- Zhiyuan Hu, Yunhai Hu, Juncheng Liu, Shuyue Stella Li, Yucheng Wang, Zhen Xu, See-Kiong Ng, Anh Tuan Luu, Xinxing Xu, Bryan Hooi, Cynthia Breazeal, and Hae Won Park. 2026. [Collaborative multi-agent test-time reinforcement learning for reasoning](#). *Preprint*, arXiv:2601.09667. ArXiv:2601.09667.
- Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2022a. Large language models can self-improve. In *EMNLP 2023*. ArXiv:2210.11610.

- Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Noah Brown, Tomas Jackson, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. 2022b. Inner monologue: Embodied reasoning through planning with language models. In *CoRL 2022*. ArXiv:2207.05608.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. [Search-r1: Training llms to reason and leverage search engines with reinforcement learning](#). *Preprint*, arXiv:2503.09516. ArXiv:2503.09516.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislaw Fort, and 17 others. 2022. [Language models \(mostly\) know what they know](#). *Preprint*, arXiv:2207.05221. ArXiv:2207.05221.
- Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. 1998. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2):99–134.
- Mark Kashirskiy and Ilya Makarov. 2026. [Sus: Strategy-aware surprise for intrinsic exploration](#). *Preprint*, arXiv:2601.10349. ArXiv:2601.10349.
- Chinmaya Kausik, Adith Swaminathan, and Nathan Kallus. 2026. [The context gathering decision process: A pomdp framework for agentic search](#). *Preprint*, arXiv:2605.07042. ArXiv:2605.07042.
- Kim et al. 2024. [Qube: Question-based belief enhancement for agentic llm reasoning](#). In *EMNLP 2024*.
- Geunwoo Kim, Pierre Baldi, and Stephen McAleer. 2023. Language models can solve computer tasks. In *NeurIPS 2023*. ArXiv:2303.17491.
- Michal Kosinski. 2023. Evaluating large language models in theory of mind tasks. In *PNAS 2024*. ArXiv:2302.02083.
- Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger. 2001. Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519.
- Shubham Kumar, Harshit Kumar, Narendra Ahuja, and Saurabh Jha. 2026. [Towards a belief-based world model for llm agents](#). *Preprint*, arXiv:2609.00455. ArXiv:2609.00455.
- Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. 2025. LLMs get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*.
- Chia-Hsuan Lee, Hao Cheng, and Mari Ostendorf. 2021. Dialogue state tracking with a language model using schema-driven prompting. In *EMNLP 2021*. ArXiv:2109.07506.
- Haohang Li, Yupeng Cao, Yangyang Yu, Shashidhar Reddy Javaji, Zhiyang Deng, Yueru He, et al. 2024. InvestorBench: A benchmark for financial decision-making tasks with LLM-based agents. *arXiv preprint arXiv:2412.18174*.
- Pengcheng Li, Zhengyang Zhang, Dongxu Zhang, Sui Huang, and Shaohua Ma. 2026a. [Vict: Verifier-instrumented credit tracing for long-horizon llm agent reinforcement learning](#). *Preprint*, arXiv:2608.28128. ArXiv:2608.28128.
- Xiaozhe Li, Tianyi Lyu, Yang Li, Yichuan Ma, Peiji Li, Linyang Li, Qipeng Guo, Dahua Lin, and Kai Chen. 2026b. [What and when to distill: Selective hindsight distillation for multi-turn agents](#). *Preprint*, arXiv:2605.19447. ArXiv:2605.19447.
- Yang Li, Yangyang Yu, Haohang Li, Zhi Chen, and Khaldoun Khashanah. 2023. [Tradinggpt: Multi-agent system with layered memory and distinct characters for enhanced financial trading performance](#). *Preprint*, arXiv:2309.03736. ArXiv:2309.03736.
- Yixia Li, Hongru Wang, Peng Lai, Zhiwen Ruan, He Zhu, Youxin Zhu, Ganlong Zhao, Minda Hu, Yun Chen, Sibe Yang, Peng Li, Jeff Z. Pan, Jia Pan, Guanhua Chen, Yang Liu, and Guanbin Li. 2026c. [Bridging the agent-world gap: Text world models for llm-based agents](#). *Preprint*, arXiv:2606.09032. ArXiv:2606.09032.
- Yuanfan Li, Qi Zhou, Wenjing Duan, and Lu Chen. 2026d. [When denser credit is not enough: Evidence-calibrated policy optimization for long-horizon llm agent training](#). *Preprint*, arXiv:2606.05885. ArXiv:2606.05885.
- Junfeng Liao, Qizhou Wang, Jianing Zhu, Bo Du, Rui Yan, and Xiuying Chen. 2026a. [Belief memory: Agent memory under partial observability](#). *Preprint*, arXiv:2605.05583. ArXiv:2605.05583.
- Sicong Liao, Zhi Chen, and Yaohua Tang. 2026b. [Tcpo: Turn-level credit policy optimization](#). *Preprint*, arXiv:2608.01667. ArXiv:2608.01667.
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let’s verify step by step. In *International Conference on Learning Representations*.
- Hongxiang Lin, Zhirui Kuai, Erpeng Xue, and Lei Wang. 2026a. [Skillc: Learning autonomous skill internalization in llm agents via contrastive credit assignment](#). *Preprint*, arXiv:2605.27899. ArXiv:2605.27899.

- Moule Lin, Qizhen Lan, Shuhao Guan, Weipeng Jing, Jiexin Fan, David Gregg, and Goetz Botterweck. 2026b. [Uncertainty quantification for llm agents: A taxonomy, an evaluation protocol, and an empirical study](#). *Preprint*, arXiv:2609.07395. ArXiv:2609.07395.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Teaching models to express their uncertainty in words. In *TMLR 2022*. ArXiv:2205.14334.
- Dongyi Liu, Yifan Niu, Qinwen Wang, Han Xiao, and Jia Li. 2026a. [Pica: Pivot-based credit assignment for search agentic reinforcement learning](#). *Preprint*, arXiv:2605.09287. ArXiv:2605.09287.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2023. Agentbench: Evaluating llms as agents. In *ICLR 2024*. ArXiv:2308.03688.
- Yicheng Liu, Bolin Zhang, Weiran Liu, Yakun Zhang, Yangqin Jiang, Zhiying Tu, and Dianhui Chu. 2026b. [Made: Belief-driven dual-agent coordination for autonomous model deployment](#). *Preprint*, arXiv:2608.01189. ArXiv:2608.01189.
- Hans-Andrea Loeliger. 2004. An introduction to factor graphs. *IEEE Signal Processing Magazine*, 21(1):28–41.
- Jia Luo. 2026. [From topology to trajectory: Llm-driven world models for supply chain resilience](#). *Preprint*, arXiv:2604.11041. ArXiv:2604.11041.
- Jinghao Luo, Yuchen Tian, Chuxue Cao, Ziyang Luo, Hongzhan Lin, Kaixin Li, Chuyi Kong, Ruichao Yang, and Jing Ma. 2026a. [From storage to experience: A survey on the evolution of llm agent memory mechanisms](#). *Preprint*, arXiv:2605.06716. ArXiv:2605.06716.
- Yu Luo, Rongchen Gao, Lu Teng, Xidao Wen, Jiamin Jiang, Qingliang Zhang, Yongqian Sun, Shenglin Zhang, Jiasong Feng, Tong Liu, Wenjie Zhang, and Dan Pei. 2026b. [Graph of states: Solving abductive tasks with large language models](#). *Preprint*, arXiv:2603.21250. ArXiv:2603.21250.
- Yijun Ma, Zehong Wang, Yiyang Li, Ziming Li, Xiaoguang Guo, Weixiang Sun, Chuxu Zhang, and Yanfang Ye. 2026. [Proplay: Procedural world models for self-evolving llm agents](#). *Preprint*, arXiv:2606.12780. ArXiv:2606.12780.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. In *NeurIPS 2023*. ArXiv:2303.17651.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of llm agents. In *ACL 2024*. ArXiv:2402.17753.
- Wachiravit Modecrua, Krittanon Kaewtawee, Krittin Pachtrachai, and Touchapon Kraisingkorn. 2026. [Multi-turn reinforcement learning for tool-calling agents with iterative reward calibration](#). *Preprint*, arXiv:2604.02869. ArXiv:2604.02869.
- Kevin Murphy. 2026. [Agentic forecasting using sequential bayesian updating of linguistic beliefs](#). *Preprint*, arXiv:2604.18576. ArXiv:2604.18576.
- Praveen Kumar Myakala, Manan Agrawal, and Rahul Manche. 2026. [Beliefshift: Benchmarking temporal belief consistency and opinion drift in llm agents](#). *Preprint*, arXiv:2603.23848. ArXiv:2603.23848.
- Kazuki Nakayashiki. 2026. [When stale constraints go unchecked: Budgeted verification failures in inherited agent memory](#). *Preprint*, arXiv:2608.25553. ArXiv:2608.25553.
- Thuy Ngoc Nguyen, Duy Nhat Phan, and Cleotilde Gonzalez. 2026. [Dynamic theory of mind as a temporal memory problem: Evidence from large language models](#). *Preprint*, arXiv:2603.14646. ArXiv:2603.14646.
- Xuying Ning, Dongqi Fu, Tianxin Wei, Hanqing Zeng, Yuanchen Bei, Bingxuan Li, Zihao Li, Qifan Wang, Xiang Shen, Yifan Wu, Jiayi Liu, Hong Li, Yinglong Xia, Xiangjun Fan, Hanghang Tong, and Jingrui He. 2026. [Evoharness-rl: Learning self-evolving runtime harness for long-horizon llm agents](#). *Preprint*, arXiv:2608.05446. ArXiv:2608.05446.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*.
- Dheeraj Mohandas Pai and Lu Xian. 2026. [Long-horizon state tracking in llms: Executing md5 through a deep sequence of dependent tool calls](#). *Preprint*, arXiv:2609.00012. ArXiv:2609.00012.
- Yijun Pan, Yukun Lian, Kunyu Shi, Junbo Li, Hongwei Xue, Sicong Xie, Guannan Zhang, and Xiaoying Xing. 2026. [Business arena: Benchmarking llm agents in a realistic marketplace](#). *Preprint*, arXiv:2608.08621. ArXiv:2608.08621.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents:

- Interactive simulacra of human behavior. In *Proceedings of the ACM Symposium on User Interface Software and Technology*.
- Jiangweizhi Peng, Yuanxin Liu, Ruida Zhou, Charles Fleming, Zhaoran Wang, Alfredo Garcia, and Mingyi Hong. 2026. [HiPER: Hierarchical reinforcement learning with explicit credit assignment for large language model agents](#). *arXiv preprint arXiv:2602.16165*.
- Liming Pu, Xiaoxia Li, Yifu Liu, Teng Cao, and Bin Yang. 2026. [Explore more, drift less: Outcome-only reinforcement learning can suffice for long-horizon interactive agents](#). *Preprint*, arXiv:2609.01245. ArXiv:2609.01245.
- Pranav Putta, Edmund Mills, Naman Garg, Sumeet Motwani, Chelsea Finn, Divyansh Garg, and Rafael Rafailov. 2024. [Agent q: Advanced reasoning and learning for autonomous ai agents](#). *Preprint*, arXiv:2408.07199. ArXiv:2408.07199.
- Qiuyi Qi, Tian Liang, Mutian Bao, Jinjian Zhang, Dongnan Liu, Wei Zhou, Linjian Mo, Ming Kong, Jie Liu, Feng Zhang, and Qiang Zhu. 2026. [Stapo: Selective trajectory-aware policy optimization for llm agent training](#). *Preprint*, arXiv:2607.04963. ArXiv:2607.04963.
- Zehan Qi, Xiao Liu, Iat Long Iong, Hanyu Lai, Xueqiao Sun, Wenyi Zhao, Yu Yang, Xinyue Yang, Jidai Sun, Shuntian Yao, Tianjie Zhang, Wei Xu, Jie Tang, and Yuxiao Dong. 2024. [Webrl: Training llm web agents via self-evolving online curriculum reinforcement learning](#). In *ICLR 2025*. ArXiv:2411.02337.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. [Direct preference optimization: Your language model is secretly a reward model](#). In *Advances in Neural Information Processing Systems*.
- Bo Ren, Yirong Mao, Yi Yang, and Wenhui Que. 2026. [Iapo: Influence-aware policy optimization for credit assignment in multi-turn service agents](#). *Preprint*, arXiv:2608.24588. ArXiv:2608.24588.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. [Proximal policy optimization algorithms](#). *arXiv preprint arXiv:1707.06347*.
- SeungWon Seo, SooBin Lim, SeongRae Noh, Haneul Kim, and HyeongYeop Kang. 2026. [From assumptions to actions: Turning llm reasoning into uncertainty-aware planning for embodied agents](#). *Preprint*, arXiv:2602.04326. ArXiv:2602.04326.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, et al. 2024. [DeepSeek-Math: Pushing the limits of mathematical reasoning in open language models](#). *arXiv preprint arXiv:2402.03300*.
- Junhong Shen, Hao Bai, Lunjun Zhang, Yifei Zhou, Amrith Setlur, Shengbang Tong, Diego Caples, Nan Jiang, Tong Zhang, Ameet Talwalkar, and Aviral Kumar. 2025. [Thinking vs. doing: Agents that reason by scaling test-time interaction](#). *Preprint*, arXiv:2506.07976. ArXiv:2506.07976.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. [Reflexion: Language agents with verbal reinforcement learning](#). In *Advances in Neural Information Processing Systems*.
- Mohit Shridhar, Xingdi Yuan, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2020. [Alf-world: Aligning text and embodied environments for interactive learning](#). In *ICLR 2021*. ArXiv:2010.03768.
- Joykirat Singh, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, Hyunji Lee, Elias Stengel-Eskin, and Mohit Bansal. 2026. [Agent-brace: Decoupling beliefs from actions in long-horizon tasks via verbalized state uncertainty](#). *Preprint*, arXiv:2605.11436. ArXiv:2605.11436.
- Pranav Singh. 2026. [When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense](#). *Preprint*, arXiv:2606.22030. ArXiv:2606.22030.
- Valentin Six, Frederik Panse, and Mathis Fajeau. 2026. [Learning pomdp world models from observations with language-model priors](#). *Preprint*, arXiv:2605.13740. ArXiv:2605.13740.
- Xinyuan Song and Zekun Cai. 2026a. [Agent vs. parametric world models: Hybrid planning for reliable language agents](#). *Preprint*, arXiv:2606.27806. ArXiv:2606.27806.
- Xinyuan Song and Zekun Cai. 2026b. [Repair the amplifier, not the symptom: Stable world-model correction for agent rollouts](#). *Preprint*, arXiv:2607.01767. ArXiv:2607.01767.
- Yuru Song and Qi Xin. 2026. [D-mem: Dopamine-gated agentic memory via reward prediction error routing](#). *Preprint*, arXiv:2603.14597. ArXiv:2603.14597.
- Tommaso Soru. 2026. [Semantic bayesian world models](#). *Preprint*, arXiv:2609.03834. ArXiv:2609.03834.
- Saksham Sahai Srivastava. 2026. [Causal intervention-based memory selection for long-horizon llm agents](#). *Preprint*, arXiv:2605.17641. ArXiv:2605.17641.
- Sukesh Subaharan. 2026. [Controlling long-horizon behavior in language model agents with explicit state dynamics](#). *Preprint*, arXiv:2601.16087. ArXiv:2601.16087.

- Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2023. Cognitive architectures for language agents. In *TMLR 2024*. ArXiv:2309.02427.
- Haofei Sun and Lin He. 2026. [When memory updates but behavior does not: Repairing implicit stale dependencies in personalized agent responses](#). *Preprint*, arXiv:2608.01619. ArXiv:2608.01619.
- Hui-Ze Tan, Xiao-Wen Yang, Hao Chen, Jie-Jing Shao, Yi Wen, Yuteng Shen, Weihong Luo, Xiku Du, Lan-Zhe Guo, and Yu-Feng Li. 2026. [Hindsight credit assignment for long-horizon llm agents](#). *Preprint*, arXiv:2603.08754. ArXiv:2603.08754.
- Sebastian Thrun, Wolfram Burgard, and Dieter Fox. 2005. *Probabilistic Robotics*. MIT Press.
- Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. 2022. Planbench: An extensible benchmark for evaluating large language models on planning and reasoning about change. In *NeurIPS 2023 Datasets and Benchmarks*. ArXiv:2206.10498.
- Daoyu Wang, Qingchuan Li, Mingyue Cheng, Jie Ouyang, Shuo Yu, Chunli Liu, Shijin Wang, Qi Liu, and Enhong Chen. 2026a. [Capo: Critic-guided action-aligned policy optimization for advancing llm agent capabilities](#). *Preprint*, arXiv:2604.18401. ArXiv:2604.18401.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023a. Voyager: An open-ended embodied agent with large language models. In *TMLR 2024*. ArXiv:2305.16291.
- Haixin Wang, Hejie Cui, Chenwei Zhang, Xin Liu, Shuowei Jin, Shijie Geng, Xinyang Zhang, Nasser Zalmout, Zhenyu Shi, and Yizhou Sun. 2026b. [T²po: Uncertainty-guided exploration control for stable multi-turn agentic reinforcement learning](#). *Preprint*, arXiv:2605.02178. ArXiv:2605.02178.
- Kevin Wang, Anna Thöni, Benjamin Kempinski, et al. 2026c. [Mindgames: A live arena for evaluating social and strategic reasoning in multi-agent llms](#). *Preprint*, arXiv:2605.29512. ArXiv:2605.29512.
- Li Wang, Yandong Wang, Xin Yu, Kui Zhang, Tianhao Peng, and Wenjun Wu. 2026d. [Tamtrl: Teacher-aligned reward reshaping for multi-turn reinforcement learning in long-context compression](#). *Preprint*, arXiv:2603.21663. ArXiv:2603.21663.
- Prince Zizhuang Wang and Shuli Jiang. 2026. [Slearl: Step-level experience augmented reinforcement learning for multi-turn agentic training](#). *Preprint*, arXiv:2603.18079. ArXiv:2603.18079.
- Xinyu Wang, Hanwei Wu, Jingwei Song, Shuyuan Zhang, Jiayi Zhang, Fanqi Kong, Tung Sum Thomas Kwok, Xiao-Wen Chang, Yuyu Luo, Chenglin Wu, and Bang Liu. 2026e. [Co-evolution of policy and internal reward for language agents](#). *Preprint*, arXiv:2604.03098. ArXiv:2604.03098.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. In *ICLR 2023*. ArXiv:2203.11171.
- Yunan Wang, Minghui Song, Zihan Zhang, Shaohan Huang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. 2026f. [Group-graph policy optimization for long-horizon agentic reinforcement learning](#). *Preprint*, arXiv:2606.22995. ArXiv:2606.22995.
- Zechuan Wang, Siyuan Lu, Hongxuan Zhang, Linjian Mo, Chenyi Zhuang, and Leilei Gan. 2026g. [Teach the magnitude, not the direction: Verifier-bounded credit assignment for multi-turn multi-step llm agents](#). *Preprint*, arXiv:2608.13179. ArXiv:2608.13179.
- Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gottlieb, Yiping Lu, Kyunghyun Cho, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, and Manling Li. 2025. [Ragen: Understanding self-evolution in llm agents via multi-turn reinforcement learning](#). *Preprint*, arXiv:2504.20073. ArXiv:2504.20073.
- Zihao Wang, Shaofei Cai, Guanzhou Chen, Anji Liu, Xiaojian Ma, and Yitao Liang. 2023b. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. In *NeurIPS 2023*. ArXiv:2302.01560.
- Ziming Wang. 2026. [Toki: A bitemporal operator algebra for contradiction resolution in llm-agent persistent memory](#). *Preprint*, arXiv:2606.06240. ArXiv:2606.06240.
- Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. 2024. [Agent workflow memory](#). *Preprint*, arXiv:2409.07429. ArXiv:2409.07429.
- Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022a. Finetuned language models are zero-shot learners. In *International Conference on Learning Representations*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022b. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS 2022*. ArXiv:2201.11903.
- Xiaobao Wu. 2025. Sailing by the stars: A survey on reward models and learning strategies for learning from rewards. *arXiv preprint arXiv:2505.02686*. Findings of EMNLP 2025.

- Xiaojun Wu, Cehao Yang, Honghao Liu, Xueyuan Lin, Wenjie Zhang, Zhichao Shi, Xuhui Jiang, Chengjin Xu, Jia Li, and Jian Guo. 2026a. [Bayesian-agent: Posterior-guided skill evolution across llm agent harnesses](#). *Preprint*, arXiv:2606.08348. ArXiv:2606.08348.
- Xuening Wu, Shan Yu, Qianya Xu, and Shenqin Yin. 2026b. [Bayesevolve: Explicit belief states for autonomous scientific discovery](#). *Preprint*, arXiv:2606.30335. ArXiv:2606.30335.
- Yue Wu, Xuan Tang, Tom M. Mitchell, and Yuanzhi Li. 2023. Smartplay: A benchmark for llms as intelligent agents. In *ICLR 2024*. ArXiv:2310.01557.
- Zhiheng Xi, Yiwen Ding, Wenxiang Chen, Boyang Hong, Honglin Guo, Junzhe Wang, Dingwen Yang, Chenyang Liao, Xin Guo, Wei He, Songyang Gao, Lu Chen, Rui Zheng, Yicheng Zou, Tao Gui, Qi Zhang, Xipeng Qiu, Xuanjing Huang, Zuxuan Wu, and Yu-Gang Jiang. 2024. [Agentgym: Evolving large language model-based agents across diverse environments](#). *Preprint*, arXiv:2406.04151. ArXiv:2406.04151.
- Zhiqiu Xia, Jinxuan Xu, Yuqian Zhang, and Hang Liu. 2025. A survey of uncertainty estimation methods on large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Yijia Xiao, Edward Sun, Di Luo, and Wei Wang. 2024. [Tradingagents: Multi-agents llm financial trading framework](#). *Preprint*, arXiv:2412.20138. ArXiv:2412.20138.
- Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, Yueru He, et al. 2024. FinBen: A holistic financial benchmark for large language models. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*.
- Yutao Xie, Nathaniel Thomas, Nicklas Hansen, Yang Fu, Li Erran Li, and Xiaolong Wang. 2026. [Tips: Turn-level information-potential reward shaping for search-augmented llms](#). *Preprint*, arXiv:2603.22293. ArXiv:2603.22293.
- Duo Xu and Faramarz Fekri. 2026. [Nesyfs: A neuro-symbolic fast-slow thinking framework for llm agent under partial observability](#). *Preprint*, arXiv:2607.28942. ArXiv:2607.28942.
- Guodong Xu. 2026. [Governed persistent memory: Source-bound state semantics and fail-closed release for long-horizon agents](#). *Preprint*, arXiv:2608.12476. ArXiv:2608.12476.
- Haoming Xu, Weihong Xu, Zongrui Li, Mengru Wang, Yunzhi Yao, Chiyu Wu, Jin Shang, Yu Gong, and Shumin Deng. 2026a. [When should models change their minds? contextual belief management in large language models](#). *Preprint*, arXiv:2605.30219. ArXiv:2605.30219.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025a. [A-MEM: Agentic memory for LLM agents](#). *arXiv preprint arXiv:2502.12110*.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025b. [A-mem: Agentic memory for llm agents](#). In *NeurIPS 2025*. ArXiv:2502.12110.
- Wujiang Xu, Yu Wang, Kai Mei, Kaiqu Liang, Zhenning Wang, Mingyu Jin, Han Zhang, Shi-Xiong Zhang, Wenyue Hua, Sambit Sahu, and Dimitris N. Metaxas. 2026b. [Memgym: a long-horizon memory environment for llm agents](#). *Preprint*, arXiv:2605.20833. ArXiv:2605.20833.
- Xiaogang Xu, Jiaqi Tang, Jianmin Chen, Yingying Yan, Zhenchao Tang, Xiangxin Zhou, Xiaobin Hu, Wei Wei, Jinfeng Wu, Qifeng Chen, Lu Zhou, Jiafei Wu, Zhe Liu, Jianwei Yin, and Weimin Zheng. 2026c. [Llm-based agents for forecasting and prediction: Methods, training, evaluation, and applications](#). *Preprint*, arXiv:2608.23058. ArXiv:2608.23058.
- Weicheng Xue. 2026. [Representation signatures and risk-feedback alignment in llm trading agents](#). *Preprint*, arXiv:2605.28850. ArXiv:2605.28850.
- Sikuan Yan, Ahmed Bahloul, Ercong Nie, Susanna Schwarzmann, Riccardo Trivisonno, Volker Tresp, and Yunpu Ma. 2026a. [Memory-r2: Fair credit assignment for long-horizon memory-augmented llm agents](#). *Preprint*, arXiv:2605.21768. ArXiv:2605.21768.
- Yandong Yan, Junwei Peng, Shijie Li, Chenxi Li, Yifei Shang, Can Deng, Ruiting Dai, Yongqiang Zhao, Jiaqi Zhu, and Yu Huang. 2026b. [Denoise-flow: Uncertainty-aware denoising for reliable llm agentic workflows](#). *Preprint*, arXiv:2603.00532. ArXiv:2603.00532.
- Jingbo Yang, Kwei-Herng Lai, Xiaowen Wang, Shiyu Chang, Yaar Harari, and Evgeniy Gabrilovich. 2026a. [Groupmembench: Benchmarking llm agent memory in multi-party conversations](#). *Preprint*, arXiv:2605.14498. ArXiv:2605.14498.
- Joshua C. Yang, Maurice Flechtner, Damian Dailisan, and Michiel A. Bakker. 2026b. [Belief engine: Configurable and inspectable stance dynamics in multi-agent llm deliberation](#). *Preprint*, arXiv:2605.15343. ArXiv:2605.15343.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. [tau-bench: A benchmark for tool-agent-user interaction in real-world domains](#). *Preprint*, arXiv:2406.12045. ArXiv:2406.12045.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS 2023*. ArXiv:2305.10601.

- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*.
- Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel, Yilun Zhao, Roy Bar-Haim, Arman Cohan, and Michal Shmueli-Scheuer. 2026. Survey on evaluation of LLM-based agents. In *Findings of the Association for Computational Linguistics: ACL 2026*. ArXiv:2503.16416.
- Hongli Yu, Tinghong Chen, Jiangtao Feng, Jiangjie Chen, Weinan Dai, Qiyang Yu, Ya-Qin Zhang, Wei-Ying Ma, Jingjing Liu, Mingxuan Wang, and Hao Zhou. 2025. Memagent: Reshaping long-context llm with multi-conv rl-based memory agent. In *ICLR 2026*. ArXiv:2507.02259.
- Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2023. Finmem: A performance-enhanced llm trading agent with layered memory and character design. In *AAAI Spring Symposium 2024*. ArXiv:2311.13743.
- Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yupeng Cao, Zhi Chen, Jordan W. Suchow, Rong Liu, Zhenyu Cui, Zhaozhuo Xu, Denghui Zhang, Koduvayur Subbalakshmi, Guojun Xiong, Yueru He, Jimin Huang, Dong Li, and Qianqian Xie. 2024. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. In *NeurIPS 2024*. ArXiv:2407.06567.
- Wenhao Yuan, Chenchen Lin, Jian Chen, Jinfeng Xu, Xuehe Wang, and Edith Cheuk Han Ngai. 2026. Verify before you commit: Towards faithful reasoning in llm agents via self-auditing. *Preprint*, arXiv:2604.08401. ArXiv:2604.08401.
- Aohan Zeng, Mingdao Liu, Rui Lu, Bowen Wang, Xiao Liu, Yuxiao Dong, and Jie Tang. 2023. Agenttuning: Enabling generalized agent abilities for llms. In *ACL 2024 Findings*. ArXiv:2310.12823.
- Chenchen Zhang. 2026. From reasoning to agentic: Credit assignment in reinforcement learning for large language models. *Preprint*, arXiv:2604.09459. ArXiv:2604.09459.
- Guibin Zhang, Hejia Geng, Xiaohang Yu, Zhenfei Yin, Zaibin Zhang, Zelin Tan, Heng Zhou, Zhongzhi Li, Xiangyuan Xue, Yijiang Li, Yifan Zhou, Yang Chen, Chen Zhang, Yutao Fan, Zihu Wang, Songtao Huang, Francisco Piedrahita-Velez, Yue Liao, Hongru Wang, and 6 others. 2025. The landscape of agentic reinforcement learning for llms: A survey. In *TMLR*. ArXiv:2509.02547.
- Linghua Zhang, Jun Wang, Jingtong Wu, and Zhisong Zhang. 2026a. Retailbench: Evaluating long-horizon autonomous decision-making and strategy stability of llm agents in realistic retail environments. *Preprint*, arXiv:2603.16453. ArXiv:2603.16453.
- Naifan Zhang, Ruihan Sun, Jinwei Su, Hengjie Yang, Zhengyuan Pan, Zhaohan Chen, and Xiaofan Zhang. 2026b. Mica: Multi-granularity intertemporal credit assignment for long-horizon emotional support dialogue. *Preprint*, arXiv:2603.06194. ArXiv:2603.06194.
- Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, Longtao Zheng, Xinrun Wang, and Bo An. 2024a. A multi-modal foundation agent for financial trading: Tool-augmented, diversified, and generalist. In *KDD 2024*. ArXiv:2402.18485.
- Xuan Zhang, Wenxuan Zhang, See-Kiong Ng, and Yang Deng. 2026c. Self-evolving world models for llm agent planning. *Preprint*, arXiv:2606.30639. ArXiv:2606.30639.
- Xuan Zhang, Zhijian Zhou, Lingfeng Qiao, Yulei Qin, Ke Li, Xing Sun, Xiaoyu Tan, Chao Qu, and Yuan Qi. 2026d. Internalizing the future: A unified agentic training paradigm for world model planning. *Preprint*, arXiv:2606.27483. ArXiv:2606.27483.
- Yudi Zhang, Meng Fang, Zhenfang Chen, and Mykola Pechenizkiy. 2026e. Self-evolving llm agents with in-distribution optimization. *Preprint*, arXiv:2606.07367. ArXiv:2606.07367.
- Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Jirong Wen. 2024b. A survey on the memory mechanism of large language model based agents. *arXiv preprint arXiv:2404.13501*.
- Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2023a. Expel: Llm agents are experiential learners. In *AAAI 2024*. ArXiv:2308.10144.
- Haotian Zhao, Songlin Zhou, Yuxin Zhang, Stephen S. T. Yau, Wenyu Zhang, Lun Tian, Tianshu Zhu, Yifeng Huang, Yucheng Zeng, Jingnan Gu, Daxiang Dong, and Jianmin Wu. 2026a. Aem: Adaptive entropy modulation for multi-turn agentic reinforcement learning. *Preprint*, arXiv:2605.00425. ArXiv:2605.00425.
- Qing Zhao, Haowei Li, Weijian Deng, Pengxu Wei, and Liang Lin. 2026b. Evoscm: Scientific belief revision through causal model evolution and experimentation. *Preprint*, arXiv:2609.01526. ArXiv:2609.01526.
- Zirui Zhao, Wee Sun Lee, and David Hsu. 2023b. Large language models as commonsense knowledge for large-scale task planning. In *NeurIPS 2023*. ArXiv:2305.14078.
- Congmin Zheng, Jiachen Zhu, Zhuoying Ou, Yuxiang Chen, Kangning Zhang, Rong Shan, Zeyu Zheng, Mengyue Yang, Jianghao Lin, Yong Yu, and Weinan Zhang. 2025. A survey of process reward

models: From outcome signals to process supervisions for large language models. *arXiv preprint arXiv:2510.08049*.

Yuyao Zheng, Haipeng Sun, Junwei Bao, Lemao Liu, Hongfei Jiang, Yang Song, and Dejing Dou. 2026. [Pgpo: Potential-guided policy optimization for multi-turn agentic tasks](#). *Preprint*, arXiv:2609.02236. ArXiv:2609.02236.

Wanjuan Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2023. Memorybank: Enhancing large language models with long-term memory. In *AAAI 2024*. ArXiv:2305.10250.

Wanjuan Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. 2023a. Language agent tree search unifies reasoning acting and planning in language models. In *ICML 2024*. ArXiv:2310.04406.

Huichi Zhou, Yihang Chen, Siyuan Guo, Xue Yan, Kin Hei Lee, Zihan Wang, Ka Yiu Lee, Guchun Zhang, Kun Shao, Linyi Yang, and Jun Wang. 2025a. [Memento: Fine-tuning llm agents without fine-tuning llms](#). *Preprint*, arXiv:2508.16153. ArXiv:2508.16153.

Pei Zhou, Aman Madaan, Srividya Pranavi Potharaju, Aditya Gupta, Kevin R. McKee, Ari Holtzman, Jay Pujara, Xiang Ren, Swaroop Mishra, Aida Nematzadeh, Shyam Upadhyay, and Manaal Faruqi. 2023b. How far are large language models from agents with theory-of-mind? In *ICLR 2024*. ArXiv:2310.03051.

Runlong Zhou, Simon S. Du, and Beibin Li. 2024a. Reflect-rl: Two-player online rl fine-tuning for lms. In *ACL 2024*. ArXiv:2402.12621.

Wangchunshu Zhou, Yuchen Eleanor Jiang, Peng Cui, Tiannan Wang, Zhenxin Xiao, Yifan Hou, Ryan Cotterell, and Mrinmaya Sachan. 2023c. [Recurrent-gpt: Interactive generation of \(arbitrarily\) long text](#). *Preprint*, arXiv:2305.13304. ArXiv:2305.13304.

Yifei Zhou, Song Jiang, Yuandong Tian, Jason Weston, Sergey Levine, Sainbayar Sukhbaatar, and Xian Li. 2025b. [Sweet-rl: Training multi-turn llm agents on collaborative reasoning tasks](#). *Preprint*, arXiv:2503.15478. ArXiv:2503.15478.

Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. 2024b. Archer: Training language model agents via hierarchical multi-turn rl. In *ICML 2024*. ArXiv:2402.19446.

Zijian Zhou, Ao Qu, Zhaoxuan Wu, Sunghwan Kim, Alok Prakash, Daniela Rus, Jinhua Zhao, Bryan Kian Hsiang Low, and Paul Pu Liang. 2025c.

[Mem1: Learning to synergize memory and reasoning for efficient long-horizon agents](#). *Preprint*, arXiv:2506.15841. ArXiv:2506.15841.

A Paper Collection Protocol

The search is scripted and rerunnable; the configuration, raw hits, and per-stage counts for the runs reported here (identifiers 20260915_v1 for 2025–2026 and 20260916_v3 for the 2020–2026 window) are released with the paper.

Queries. Twenty-three keyword queries were defined in the first run, four to five per survey section; a second run added eleven queries on evidence acquisition and theory-of-mind belief modelling (34 in total): representation (Section 4), revision (Section 5), learning (Section 6), evidence (Section 7), and evaluation (Section 8).

Sources. Source A queried OpenAlex across all venues with publication date from 1 January 2025, 50 results per query. Source B queried OpenAlex restricted to the arXiv source with publication date from 1 January 2026, 50 results per query, to improve recall of preprints. The arXiv API itself was rate-limited during the run; the OpenAlex mirror of arXiv lags by a few days.

Screening. Records were deduplicated by identifier and then by normalised title, so that a preprint and its journal version count once. Rule-based screening on title and abstract applied one inclusion criterion for an LLM-related term (I1), one exclusion criterion for off-domain applications (E1), and one inclusion criterion requiring at least two topic terms from a fixed list (I2). Records without an abstract of at least 200 characters were excluded. Included records were ranked by a relevance score used only for ordering the manual full-text screen, never for inclusion.

Second and third runs. The second run (34 queries, same sources and rules) returned 834 unique records, 338 after screening, of which 96 were published in 2026; only two of those had not already been read in the first run, which we take as evidence that the first run had saturated the keyword neighbourhood. The third run extended the window to 1 January 2020 with 100 results per query and returned 4,471 unique records, 3,999 after title deduplication and 2,079 after rule screening. Keyword ranking is dominated by generic high-citation papers before 2025, so the 2020–2025 foundations were enumerated as a curated

Stage	Count
Queries	23
Source A raw hits / unique	1,150 / 715
Source B raw hits / unique	1,150 / 1,008
Merged unique (identifier)	1,685
After title-level deduplication	1,460
Excluded: no or short abstract	45
Excluded I1: no LLM term	160
Excluded E1: off-domain	35
Excluded I2: fewer than two topic terms	357
Included after rule screening	863
of which published in 2026	656
Full-text screened	117
of which included after reading	106

Table 7: Paper-collection counts for run 20260915_v1.

list of 90 titles resolved through OpenAlex for venue and citation data (Table 13); 75 of them were read in full and 73 included.

Evidence level. Each cited source is labelled P (peer-reviewed venue) or A (arXiv preprint at the time of writing) in the tables and in Figure 5. Of the 179 papers included after full-text reading, 47 are peer-reviewed and 132 are preprints; every number quoted in the text is the paper’s own headline figure, taken from the abstract or the reported table, and none has been reproduced by us.

Full-text screening. Each paper read in full was recorded with the fields: who maintains the belief, credited object, revision signal, whether belief-level metrics are reported, survey section, and role. The record is the source of Tables 8–11 and Figure 5.

B Additional Tables and Figures

C Timeline of Credited Objects

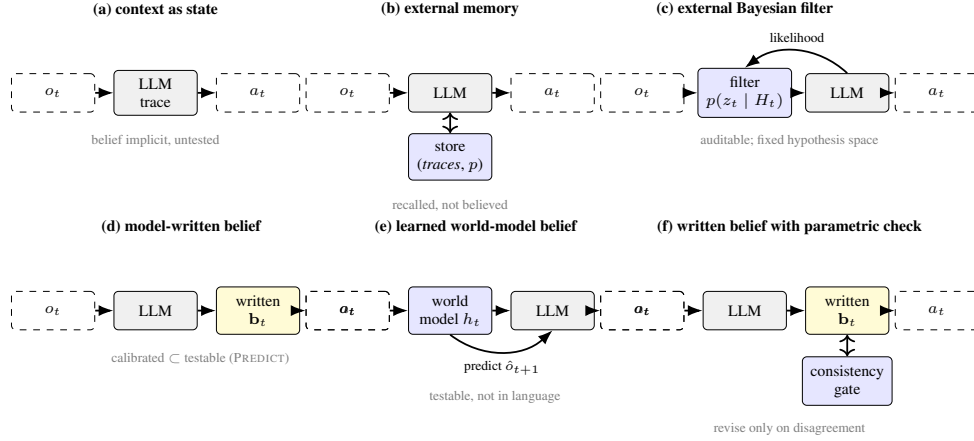


Figure 6: Who maintains the belief. Yellow marks a belief the model writes and can reason about; blue marks an external estimator. Only (c), (d)-testable, (e), and (f) commit to a prediction that later evidence can falsify.

Class	Method	Model-written	Append-only	Uncertainty	TEST	REVISE	Domain
Context as state	ReAct (Yao et al., 2023b)	✓	✓	✗	✗	✗	general
External memory	MemGPT (Packer et al., 2023)	✗	✗	✗	✗	store update	dialogue
Probabilistic memory	BeliefMem (Liao et al., 2026a)	✗	✗	prob.	✗	noisy-OR	LoCoMo, ALFWorld
Probabilistic memory	D-MEM (Song and Xin, 2026)	✗	✗	RPE	✗	gated write	LoCoMo
External filter	Belief-State Engine (Chattopadhyay and Halder, 2026)	✗	✗	posterior	✓	Bayes	Tiger, security
External filter	Belief at Risk (Dixon, 2026b)	✗	✗	posterior	✓	Bayes	equities
Model-written	Agent-BRACE (Singh et al., 2026)	✓	✗	ordinal	✗	RL	ALFWorld-style
Model-written	Linguistic forecaster (Murphy, 2026)	✓	✓	numeric	✓	sequential	ForecastBench
Model-written	BOND (Cui and Lin, 2026)	✓	✓	posterior tag	✗	distilled	negotiation
Model-written	CBM (Xu et al., 2026a)	✓	✓	✗	✓(symbolic)	RL	closed world
Model-written	EvoSCM (Zhao et al., 2026b)	✓	✓	population	✓(interventions)	rule distillation	science
World model	BB-WM (Kumar et al., 2026)	✗	✗	latent	✓	learned	general

Table 8: Belief representation by who maintains the belief. TEST: the belief commits to a falsifiable prediction. All 2026 entries are arXiv preprints at the time of writing; ReAct and MemGPT are peer-reviewed.

Method	Failed stay	Failed update	Failed isolation	Failed act	Trigger	Signal projected to
Reflexion (Shinn et al., 2023)		✓			task failure	whole context
BeliefShift (Myakala et al., 2026)	✓	✓			user evidence	(benchmark)
STALE (Chao et al., 2026)		✓		✓	implicit conflict	(benchmark)
CBM (Xu et al., 2026a)	✓	✓	✓		symbolic check	belief state
TOKI (Wang, 2026)		✓			contradiction	memory rows
MemOps (Hao et al., 2026)		✓	✓		lifecycle op	(benchmark)
D-MEM (Song and Xin, 2026)		✓	✓		reward pred. error	memory path
STOCKTAKE (Deb and Krishnan, 2026)		✓		✓	oracle mismatch	(benchmark)
Agent-BRACE (Singh et al., 2026)		✓		✓	certainty	belief model
Werewolf audit (Gao et al., 2026b)		✓		✓	external belief log	(benchmark)

Table 9: Belief revision methods and benchmarks by failure mode addressed, revision trigger, and where the signal is projected.

Level	Representative methods	Feedback source	Granularity	Signed by returned o_{t+1}	Delayed evidence	Partial obs.	Evidence
Output	SFT, RLHF, DPO (Ouyang et al., 2022; Rafailov et al., 2023)	demos, preferences	response	✗	✗	✗	P
Trajectory	PPO, GRPO, PRMs (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024)	verifier, PRM	step / rollout	✗	✗	✗	P
Interaction	HiPER, G ² PO, ECPO, IAPPO (Peng et al., 2026; Wang et al., 2026f; Li et al., 2026d; Ren et al., 2026)	environment	turn / action	✗	partial	✗	A
Interaction	Memory-R2, TAMTRL, EvoHarness-RL (Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026)	environment, teacher	memory operation	✗	partial	✓	A
State	CBM (Xu et al., 2026a)	symbolic belief reward	belief state	closed world	✓	✓	A
State	Agent-BRACE (Singh et al., 2026)	joint RL, calibration	claim + certainty	calibration target	✓	✓	A
State	BOND (Cui and Lin, 2026)	Bayesian teacher	posterior tag	teacher	✗	✓	A

Table 10: Learning signals by credited object. Evidence: P = peer-reviewed venue, A = arXiv preprint at the time of writing.

Benchmark	Domain	Horizon	Latent state	Observable to evaluator	Delayed evidence	Belief-level quantity reported
ALFWorld (Shridhar et al., 2020)	text + embodied	episodic	household state	✓	partial	none (task success)
PlanBench (Valmeekam et al., 2022)	classical planning	episodic plan-length	PDDL state	✓	✗	plan verification, replanning
AgentBench (Liu et al., 2023)	8 environments	episodic	environment state	✓	partial	none (task success)
SmartPlay (Wu et al., 2023)	6 games	episodic	game state	✓	partial	none (9 capability scores)
T4D / FaR (Zhou et al., 2023b)	theory of mind	single-turn	others' beliefs	✓(annotated)	✗	belief tracking vs. action selection
False-belief tasks (Kosinski, 2023)	theory of mind	single-turn	others' beliefs	✓(annotated)	✗	false-belief accuracy
LoCoMo (Maharana et al., 2024)	long conversations	35 sessions	conversation facts	annotated	✓	temporal QA (proxy)
7-bench / r^2 -bench (Yao et al., 2024; Barres et al., 2025)	tool-user interaction	multi-turn	database state	✓	✓	pass-% consistency; reasoning vs. communication errors
MemoryAgentBench (Hu et al., 2025a)	incremental multi-turn	long-horizon	memory contents	annotated	✓	retrieval, test-time learning, conflict resolution
BeliefTrack (Xu et al., 2026a)	closed-world reasoning	multi-turn	finite belief space	✓(symbolic)	✓	turn-level belief accuracy; failed state/update/isolation
DeltaLogic (Dhanda, 2026)	logical reasoning	2 steps	premise set	✓	✓	revision accuracy vs. initial accuracy
DTOM-Track (Nguyen et al., 2026)	dialogue	multi-turn	interlocutor belief	✓(annotated)	✓	prior-belief recall; change detection
BeliefShift (Myakala et al., 2026)	user modelling	multi-session	user opinion	annotated	✓	revision accuracy; drift coherence; evidence sensitivity
OmniTOM (Bawaneh et al., 2026)	theory of mind	narrative	actors' beliefs	✓(annotated)	✓	belief-proposition extraction; schema labels
MINDGAMES (Wang et al., 2026c)	social games	repeated games	hidden roles, opponents	✗	✗	TrueSkill; turn-level logs
MEDLEY-BENCH (Abbahi et al., 2026)	social pressure	2 rounds	—	✗	✗	sensitivity to manipulated consensus
STALE (Chao et al., 2026)	memory validity	multi-turn	fact validity	annotated	✓	state resolution; premise resistance; policy adaptation
MemOps (Hao et al., 2026)	long conversations	multi-session	memory lifecycle	annotated	✓	operation-level probes
GroupMemBench (Yang et al., 2026a)	multi-party dialogue	multi-session	per-speaker knowledge	annotated	✓	information-update accuracy
MemGym (Xu et al., 2026b)	mixed agent tasks	long-horizon	—	✗	partial	memory-isolated score delta
STOCKTAKE (Deb and Krishnan, 2026)	inventory control	weekly, long	demand factors	✓(Bayes filter)	✓	detection lag; skill vs. oracle; knowing-doing rate
Flux (Alawad et al., 2026)	text game	Reservoir	simulator state	✓	✓	state-tracking error modes
YC-Bench (He et al., 2026a)	startup simulation	hundreds of turns	market, clients	privileged outcome	✓	scrapped usage (proxy)
RetailBench (Zhang et al., 2026a)	retail POMDP	180 days	demand, supply	privileged oracle	✓	evidence-acquisition behaviour (proxy)
EnterpriseArena (Han et al., 2026)	lending	132 months	borrower pool	privileged outcome	✓	observation-budget allocation (proxy)
Business Arena (Pan et al., 2026)	marketplace	long-horizon	market	privileged outcome	✓	action-level attribution
Werewolf audit (Gao et al., 2026b)	social deduction	9-player game	roles	✓	✓	action-belief consistency
CAGE-2 study (Bogdanov et al., 2026)	cyber defence	episodic	network state	✓	partial	state-abstraction return gain
MDS state tracking (Pai and Xian, 2026)	tool chains	196 calls	registers	✓	✗	exact-state success
FinBench (time-gated) (Ghosh and Devarakonda, 2026)	asset forecasting	multi-week	returns	realised	✓	Brier; interval scores
TradingArena (Xue, 2026)	trading	multi-week	regimes	realised	✓	representation drift (early warning)
HERCULEAN (Anonymous, 2026b)	finance workflows	multi-step	market, books	partial	✓	state-tracking sub-score
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024)	finance tasks / trading	task / episodic	—	✗	partial	none (outcome only)

Table 11: Benchmarks that exercise belief maintenance, by whether the evaluator can observe the latent state and what belief-level quantity is reported. “Annotated” means human labels stand in for the latent state; “privileged” means the simulator knows it but the benchmark reports only outcomes. Ev. column omitted for space; peer-reviewed entries are ALFWorld, PlanBench, AgentBench, SmartPlay, T4D, false-belief tasks, LoCoMo, FinBen, and InvestorBench; the rest are arXiv preprints at the time of writing.

System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.	System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.
Schema-driven DST (Lee et al., 2021)	2021	written	none	dialogue history plus schema...	✓	P	Cyclic subtask graphs (Charzedeine and Saab, 2026)	2026	written	none	state-analysis-and-routing...	—	A
Verbalized Uncertainty (Lin et al., 2022)	2022	written	output	self-evaluation P and P of...	✓	A	CHAL (Giovannelli and Kent, 2026)	2026	written	none	gradient-informed revision on...	o	A
DrumV3 (Hafner et al., 2023)	2023	world model	world-model loss	RSSM latent posterior updated...	o	P	Predictor (Six et al., 2026)	2026	world model	world-model loss	belief-based likelihood score...	o	A
RecurrentGPT (Zhou et al., 2023c)	2023	written	output	LLM rewrites its own...	o	A	Belief Engine stance... (Yang et al., 2026b)	2026	world model	none	log-odds update with evidence...	✓	A
LLM-MCTS (Zhou et al., 2023b)	2023	world model	trajectory	LLM-sampled world state...	o	P	HiPF (Diao et al., 2026)	2026	written	trajectory	subgoal completion +...	o	A
RAP (Hao et al., 2023)	2023	world model	trajectory	LLM-as-world-model predicted...	o	P	ProPlay (Ma et al., 2026)	2026	world model	trajectory	post-episode graph refinement...	o	A
Reflect-RL (Zhou et al., 2024a)	2024	written	interaction	reverse reflection model...	o	P	POMDP validation... (Dixon, 2026c)	2026	ext. filter	none	Bayesian	✓	A
Hypothetical Minds (Cross et al., 2024)	2024	written	none	hypotheses about other...	o	A	Internalizing the Future (Zhang et al., 2026d)	2026	written	trajectory	foreight-conditioned RL...	o	A
Talker Reasoner (Christakopoulos and Mourad, 2024)	2024	written	none	Reasoner writes/updates...	o	A	Hybrid-WM (Song and Cai, 2026a)	2026	world model	world-model loss	consistency gate	✓	A
QUBE (Kim et al., 2024)	2024	written	none	question-answering over...	o	A	Governance-aware POMDP... (Dixon, 2026a)	2026	ext. filter	none	Bayesian update on evidence...	o	A
MEMI (Zhou et al., 2025c)	2025	written	outcome (written state)	internal state rewritten each...	o	A	BayoEvolve (Wu et al., 2026b)	2026	ext. filter	none	posterior refit after each...	✓	A
MemAgent (Yu et al., 2025)	2025	written	outcome (written state)	task-wise overwrite of...	o	A	WorldEvolve (Zhang et al., 2026c)	2026	world model	world-model loss	prediction-observation...	o	A
PCE (Seo et al., 2026)	2026	written	none	scenario likelihood scoring...	o	A	WM-SAR (Song and Cai, 2026b)	2026	world model	trajectory	spectral error amplification	o	A
RB-VLA (Bagaria et al., 2026)	2026	world model	world-model loss	self-supervised world-model...	o	A	BCO (Chen et al., 2026b)	2026	written	none	evaluation of new candidate...	✓	A
REDEREF (Hossaini et al., 2026)	2026	prob. memory	interaction	judge feedback updates...	o	A	EvosCM causal belief... (Zhao et al., 2026b)	2026	written	none	prediction-observation...	✓	A
TAMTRL (Wang et al., 2026d)	2026	written	trajectory	teacher likelihood of memory...	o	A	CAPTURE (Hossaini et al., 2026)	2026	world model	world-model loss	uncertainty-triggered...	✓	A
Self-Guide internal... (Wang et al., 2026e)	2026	written	trajectory	self-generated guidance as...	o	A	Semantic Bayesian World... (Soru, 2026)	2026	ext. filter	none	Bayesian conditioning on...	o	A
ReflectChain... (Lao, 2026)	2026	world model	trajectory	latent trajectory rehearsal...	o	A	Belief-State Engine (Chattopadhyay and Halder, 2026)	2026	ext. filter	none (prompting)	Bayesian posterior	✓	A

Table 12: Systems with an explicit belief mechanism, by year. Belief kept by: who maintains the belief (Section 4); Credited: the object that receives the learning signal under the criterion of Section 6, none if the belief is not trained; Metric: ✓ reports a belief-level quantity, o a proxy; Ev.: P peer-reviewed, A arXiv preprint.

System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.	System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.
Decision Transformer (Chen et al., 2021)	2021	context	interaction	return-to-go conditioning...		P	HCAPO hindsight critic (Tan et al., 2026)	2026	context	trajectory	post-hoc LLM critic refinement		A
Schema-driven DST (Lee et al., 2021)	2021	written	none	dialogue history plus schema...	✓	P	TIPS (Xie et al., 2026)	2026	context	interaction	teacher answer-likelihood gain	o	A
Chain-of-Thought (Wei et al., 2022b)	2022	context	output	none		P	DTOM-Track (Nguyen et al., 2026)	2026	context	none	none		A
Verbalized Uncertainty (Lin et al., 2022)	2022	written	output	fine-tuning on calibrated...	✓	P	D-MEM (Song and Xian, 2026)	2026	ext. memory	none	reward prediction error gating	o	A
LMs Know (Kadavath et al., 2022)	2022	written	output	self-evaluation P and P of...	✓	A	SLEA-RL (Wang and Jiang, 2026)	2026	ext. memory	trajectory	observation change triggers...	o	A
Inner Monologue (Huang et al., 2022b)	2022	context	interaction	success detection, scene...		P	Graph of States (Lao et al., 2026b)	2026	ext. memory	none	symbolic check	o	A
DrumV3 (Hafner et al., 2023)	2023	world model	world-model loss	RSSM latent posterior updated...	o	P	TAMTRL (Wang et al., 2026d)	2026	written	trajectory	teacher likelihood of memory...	o	A
DEPS (Wang et al., 2023b)	2023	context	trajectory	descriptor of execution...		P	MT-GRPO Iterative... (Modersena et al., 2026)	2026	context	interaction	interaction		A
Reflection (Shim et al., 2023)	2023	ext. memory	trajectory	verbal self-reflection after...		P	Self-Guide internal... (Wang et al., 2026e)	2026	written	trajectory	self-generated guidance as...	o	A
RCI (Kim et al., 2023)	2023	context	interaction	recursive self-criticism of...		P	SAVER (Yuan et al., 2026)	2026	context	none	adversarial audit localizing...	o	A
Self-Refine (Madani et al., 2023)	2023	context	output	self-generated...		P	ReflectChain... (Lao, 2026a)	2026	world model	trajectory	latent trajectory rehearsal...	o	A
Generative Agents (Park et al., 2023)	2023	ext. memory	interaction	observation stream, periodic...		P	Cyclic subtask graphs (Charzedeine and Saab, 2026)	2026	written	none	state-analysis-and-routing...	o	A
Self-Debug (Chen et al., 2023b)	2023	context	output	execution feedback +...		P	CAPO (Wang et al., 2026a)	2026	context	interaction	none		A
Tree of Thoughts (Yao et al., 2023a)	2023	context	trajectory	self-evaluation of...		P	AEM adaptive entropy... (Zhao et al., 2026a)	2026	context	trajectory	response-level surprisal x...	o	A
MemoryBank (Zhong et al., 2023)	2023	ext. memory	output	Ebbinghaus-style forgetting...		P	T2PO (Wang et al., 2026b)	2026	context	interaction	token-level...	o	A
RecurrentGPT (Zhou et al., 2023c)	2023	written	output	LLM rewrites its own...		P	MedExAgent (Gao et al., 2026a)	2026	context	output	none		A
LLM-MCTS (Zhou et al., 2023b)	2023	world model	trajectory	LLM-as-world-model state...	o	P	TRACE turn credit for... (He et al., 2026b)	2026	context	interaction	leave-one-turn-out semantic...	o	A
RAP (Hao et al., 2023)	2023	world model	trajectory	LLM-as-world-model predicted...		P	PCA post-based credit (Liu et al., 2026a)	2026	context	trajectory	prior-step potential-based...	o	A
Voyager (Wang et al., 2023a)	2023	ext. memory	trajectory	environment execution...		P	CHAL (Giovannelli and Kent, 2026)	2026	written	trajectory	gradient-informed revision on...	o	A
Expel. (Zhao et al., 2023a)	2023	ext. memory	trajectory	cross-task insight extraction...		P	Why world models (Alawad et al., 2026)	2026	context	none	none		A
CoALA (Sumers et al., 2023)	2023	ext. memory	none	conceptual		P	Predictor (Six et al., 2026)	2026	world model	world-model loss	belief-based likelihood score...	o	A
TradingGPT (Li et al., 2023)	2023	ext. memory	interaction	layered memory decay +...		A	Belief Engine stance... (Yang et al., 2026b)	2026	prob. memory	none	log-odds update with evidence...	✓	A
FaR / T4D (Zhou et al., 2023b)	2023	context	interaction	foresee-and-reflect...		P	CAGE-2 compound agent... (Bogdanov et al., 2026)	2026	ext. memory	none	deterministic state-tracking...	o	A
LATS (Zhou et al., 2023a)	2023	ext. memory	trajectory	MCTS backup of LM-valued...	o	P	Representation... (Xue, 2026)	2026	context	output	structured risk feedback	o	A
FinAct (Chen et al., 2023a)	2023	context	trajectory	supervised fine-tuning on...		A	CM1 (Srivastava, 2026)	2026	ext. memory	none	counterfactual answer-score...	o	A
MemGPT (Packer et al., 2023)	2023	ext. memory	output	self-directed memory paging...		A	SERL selective... (Li et al., 2026b)	2026	context	interaction	per-step environment feedback...	o	A
AgentTuning (Zeng et al., 2023)	2023	context	trajectory	hybrid instruction tuning on...		P	Memory-R2 / LoGo-GRPO (Yan et al., 2026a)	2026	ext. memory	none	local rollouts from shared...	o	A
FinMem (Yu et al., 2023)	2023	ext. memory	interaction	layered memory with...		P	SkiliC (Lin et al., 2026a)	2026	context	trajectory	skill-vs-no-skill reward gap	o	A
Reflect-RL (Zhou et al., 2024a)	2024	written	interaction	fraction reflection model...		P	AIK financial harness (Borjigin et al., 2026)	2026	ext. memory	none	passive knowledge injection...	o	A
FinAgent (Zhang et al., 2024a)	2024	ext. memory	interaction	low-level and high-level...		P	TOKI (Wang, 2026)	2026	ext. memory	none	binomial operators	o	A
AgentGym (Xi et al., 2024)	2024	context	trajectory	AgentEvolve		A	Q-Evolve (Zhang et al., 2026e)	2026	context	trajectory	advantage from...	o	A
Hypothetical Minds (Cross et al., 2024)	2024	written	none	hypotheses about other...		A	HiPF (Diao et al., 2026)	2026	written	trajectory	subgoal completion +...	o	A
FinCon (Yu et al., 2024)	2024	ext. memory	none	episodic self-critique...		P	ProPlay (Ma et al., 2026)	2026	world model	trajectory	post-episode graph refinement...	o	A
Agent Q (Putta et al., 2024)	2024	ext. memory	trajectory	MCTS with AI self-critique...		A	POMDP validation... (Dixon, 2026c)	2026	ext. filter	none	Bayesian	✓	A
Agent Workflow Memory (Wang et al., 2024)	2024	ext. memory	none	induction of reusable...		A	G2PO (Wang et al., 2026f)	2026	context	trajectory	foreight-conditioned TD...	o	A
Talker Reasoner (Christakopoulos and Mourad, 2024)	2024	written	none	Reasoner writes/updates...		A	Internalizing the Future (Zhang et al., 2026d)	2026	written	trajectory	foreight-conditioned RL...	o	A
QUBE (Kim et al., 2024)	2024	written	none	question-answering over...		P	Hybrid-WM (Song and Cai, 2026a)	2026	world model	world-model loss	consistency gate	✓	A
TradingAgents (Xiao et al., 2024)	2024	context	output	multi-agent debate and...		A	Governance-aware POMDP... (Dixon, 2026a)	2026	ext. filter	none	Bayesian update on evidence...	o	A
AgentRefine (Fu et al., 2025)	2025	context	trajectory	environment observation...		P	BayoEvolve (Wu et al., 2026b)	2026	ext. filter	none	posterior refit after each...	✓	A
LOOP (Chen et al., 2025a)	2025	context	trajectory	environment feedback via API...		P	WorldEvolve (Zhang et al., 2026c)	2026	world model	world-model loss	prediction-observation...	o	A
A-Mem (Xu et al., 2025b)	2025	ext. memory	output	memory evolution		P	WM-SAR (Song and Cai, 2026b)	2026	world model	trajectory	spectral error amplification	o	A
ReSearch (Chen et al., 2025b)	2025	context	output	retrieved documents...		A	STAPO... (Qi et al., 2026)	2026	context	trajectory	normalized-entropy outlier...	o	A
RAGEN / StarPO (Wang et al., 2025)	2025	context	trajectory	search results inserted into...		P	Auditing... (Gao et al., 2026b)	2026	ext. memory	none	logged per-turn belief updates	✓	A
Mem0 (Chikara et al., 2025)	2025	ext. memory	none	environment reward...		A	RLAW (Ali and Tseng, 2026)	2026	ext. memory	interaction	critique score	o	A
TTI (Shen et al., 2025)	2025	context	trajectory	LLM-driven extraction...		A	NeSyFS (Xu and Fekri, 2026)	2026	ext. memory	interaction	repeated fast-action failure...	o	A
MEMI (Zhou et al., 2025c)	2025	written	outcome (written state)	additional environment...		A	MADE (Liu et al., 2026b)	2026	ext. memory	interaction	execution failure ->...	o	A
MemAgent (Yu et al., 2025)	2025	written	outcome (written state)	chunk-wise overwrite of...		P	TPCO turn-level credit (Liao et al., 2026b)	2026	context	interaction	majority vote over arithmetic	o	A
ARPO (Dong et al., 2025a)	2025	context	trajectory	token entropy spike after...		A	When Memory Updates but... (Sun and He, 2026)	2026	ext. memory	none	state dependency repair	o	A
Memorito (Zhou et al., 2025b)	2025	ext. memory	trajectory	environment feedback, risk...		P	LAPO (Ren et al., 2026)	2026	ext. memory	none	learned...	o	A
Agent-R1 (Cheng et al., 2025)	2025	context	trajectory	tool/environment feedback per...		A	GPm (Xu, 2026)	2026	ext. memory	none	symbolic check	o	A
MATTRL test-time... (Hu et al., 2026)	2026	ext. memory	interaction	turn-level credit for...		A	CEST (Wang et al., 2026g)	2026	context	interaction	verified turn reward with...	o	A
SuS strategy-aware... (Kashirsky and Makarov, 2026)	2026	context	output	surprise residual		A	HIDITTR (Guo et al., 2026)	2026	context	interaction	group tool-call success rate	o	A
External affective... (Sobharwal, 2026)	2026	ext. memory	none	exponential smoothing or...		A	Intentional	2026	context	interaction	intentional	o	A
PCE (Seo et al., 2026)	2026	context	none	scenario likelihood scoring...		A	BCO (Chen et al., 2026b)	2026	written	none	evaluation of new candidate...	✓	A
HIFER (Peng et al., 2026)	2026	context	trajectory	self-supervised world-model...		A	EvoSCAM capture belief... (Zhao et al., 2026b)	2026	written	none	prediction-observation...	✓	A
RB-VLA (Bagaria et al., 2026)	2026	world model	world-model loss	self-supervised world-model...	o	A	CAPTURE (Hossain et al., 2026)	2026	world model	world-model loss	uncertainty-triggered...	o	A
REDDER (Hossain et al., 2026)	2026	prob. memory	interaction	judge feedback updates...		A	PPPO (Zhang et al., 2026)	2026	context	interaction	potential difference between...	o	A
BrainSolveFlow (Yan et al., 2026)	2026	ext. memory	interaction	episodic entropy over samples...		A	Semantic Bayesian World Model (Soru, 2026)	2026	ext. filter	none	Semantic Bayesian World Model...	o	A
MICA (Zhang et al., 2026b)	2026	ext. memory	interaction	none		A	Belief-State Engine (Chattopadhyay and Halker, 2026)	2026	ext. filter	none (prompting)	Bayesian posterior	o	A

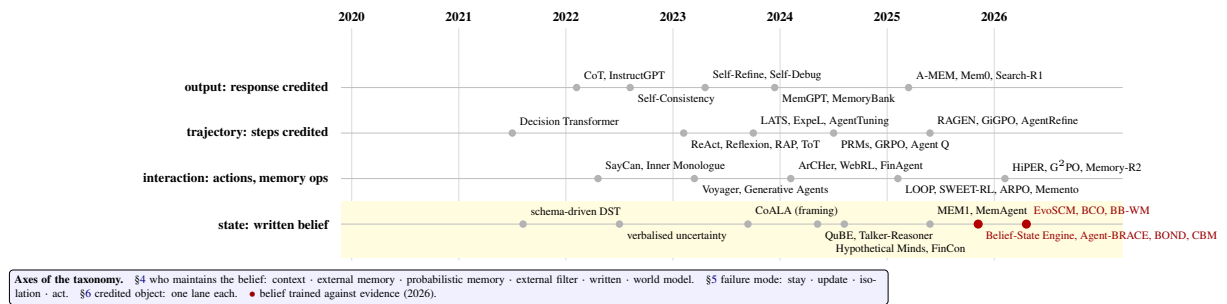


Figure 7: Where credit has gone, 2020–2026. Lanes are credited objects; Table 12 lists every screened system. Written beliefs exist from dialogue state tracking onward, but until 2026 they are credited by task outcome or not trained at all; the red points are the first to give the belief its own learning signal.